

VIRGINIA STANDARDS OF LEARNING

TEST ITEM SET

ALGEBRA II

2009 Mathematics Standards of Learning

Released Spring 2015

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SAMPLE A

Which expression is equivalent to $\sqrt{\frac{7x}{16}}$?

☐ **A** $\frac{7x}{4}$

☐ **B** $\frac{7x}{8}$

☐ **C** $\frac{\sqrt{7x}}{4}$

☐ **D** $\frac{\sqrt{7x}}{8}$

Directions: Type your answer in the box.

SAMPLE B

What value of x makes $\sqrt{x} - 3 = 6$ true?

Which expression is equivalent to $\sqrt{20x^{16}y^{25}}$ for positive x and y values?

- ☐ A $2x^4y^5\sqrt{5}$
- ☐ B $5x^4y^5\sqrt{2}$
- ☐ C $2x^8y^{12}\sqrt{5y}$
- ☐ D $5x^8y^{12}\sqrt{2y}$

Which expression is equivalent to $\sqrt[3]{6w^7} \cdot \sqrt[3]{4w^5}$?

- ☐ A $2w^4\sqrt[3]{3}$
- ☐ B $2w^4\sqrt[3]{6}$
- ☐ C $2w^{11}\sqrt[3]{3w^2}$
- ☐ D $2w^{11}\sqrt[3]{6w^2}$

The steps used to solve an equation are shown.

Step 1: $\frac{2}{3}r = 14i$

Step 2: $\left(\frac{3}{2}\right)\frac{2}{3}r = 14i\left(\frac{3}{2}\right)$

Step 3: $\left(\frac{3}{2} \cdot \frac{2}{3}\right)r = 14i\left(\frac{3}{2}\right)$

Step 4: $1 \cdot r = 21i$

Step 5: $r = 21i$

What property justifies the work between Step 4 and Step 5 ?

- ☐ A Identity property of multiplication
- ☐ B Inverse property of multiplication
- ☐ C Commutative property of multiplication
- ☐ D Associative property of multiplication

Which expression is equivalent to the following expression if no denominators equal zero?

$$\frac{\frac{11-w}{30w^2}}{\frac{w-11}{5w^6}}$$

☐ **A** $-\frac{w^4}{6}$

☐ **B** $-\frac{6}{w^3}$

☐ **C** $\frac{w^3}{6}$

☐ **D** $\frac{6}{w^4}$

What is the complete factorization of $(18x^4 + 12x^3 - 6x)$?

- ☐ A $6x^3(3x + 2)$
- ☐ B $6x(3x^3 + 2x^2)$
- ☐ C $6x(3x - 1)(x + 1)$
- ☐ D $6x(3x^3 + 2x^2 - 1)$

Which of these is equivalent to i^{75} ?

☐ A i

☐ B $-i$

☐ C 1

☐ D -1

For which value of b is $x^2 + bx - 60$ factorable over the set of integers?

- ☐ A 61
- ☐ B 23
- ☐ C -7
- ☐ D -16

If no denominator equals zero, which expression is equivalent to $\frac{25 - 4x^2}{6x^2 + 9x - 15} \cdot \frac{6x^2 - 2x - 4}{2x^2 - x - 10}$?

- ☐ A -2
- ☐ B 2
- ☐ C $\frac{-2(3x + 2)}{3(x + 2)}$
- ☐ D $\frac{2(3x + 2)}{3(x + 2)}$

Assuming the denominators do NOT equal zero, which expression is equivalent to $\frac{12}{x+1} + \frac{1}{x-4}$?

☐ A $\frac{13x - 47}{(x+1)(x-4)}$

☐ B $\frac{13}{(x+1)(x-4)}$

☐ C $\frac{13x - 47}{2x - 3}$

☐ D $\frac{13}{2x - 3}$

Which expression is equivalent to $\sqrt{36x^9y^{25}}$, where $x > 0$ and $y > 0$?

☐ A $6x^3y^5$

☐ B $6x^{\frac{9}{2}}y^{\frac{25}{2}}$

☐ C $18x^3y^5$

☐ D $18x^{\frac{9}{2}}y^{\frac{25}{2}}$

What nonzero value of x is a solution to the following equation?

$$\frac{x+2}{x} + \frac{x-6}{3x} = \frac{2x+9}{5x}$$

- ☐ A $x = \frac{27}{14}$
- ☐ B $x = \frac{17}{14}$
- ☐ C $x = \frac{13}{14}$
- ☐ D $x = \frac{5}{14}$

How many values of x will satisfy the equation $-2|3x - 5| = 0$?

☐ A 0

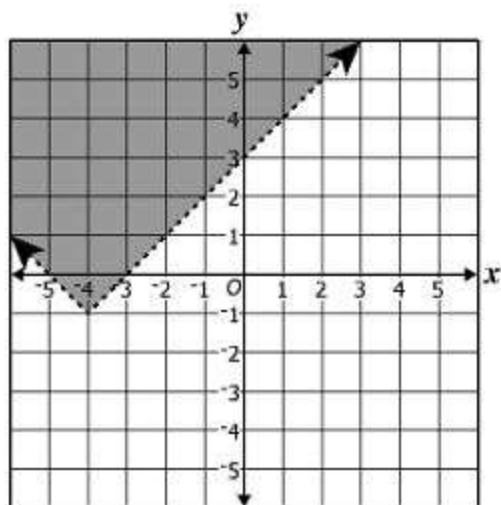
☐ B 1

☐ C 2

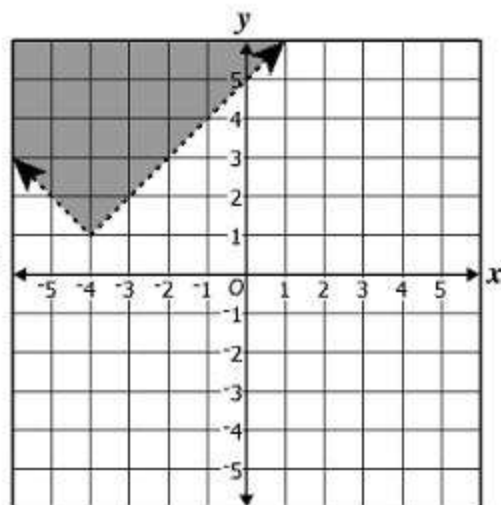
☐ D 3

Which graph best represents the solutions for $y < |x + 4| - 1$?

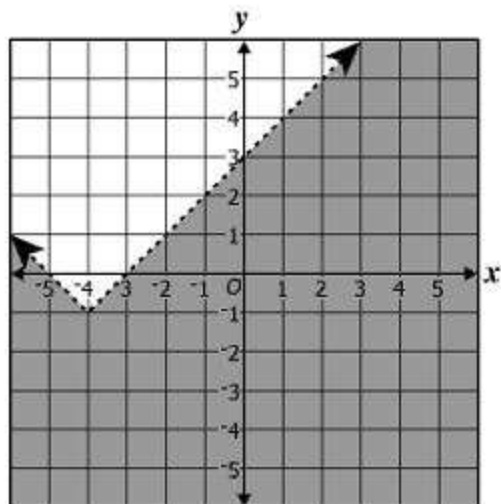
☐ A



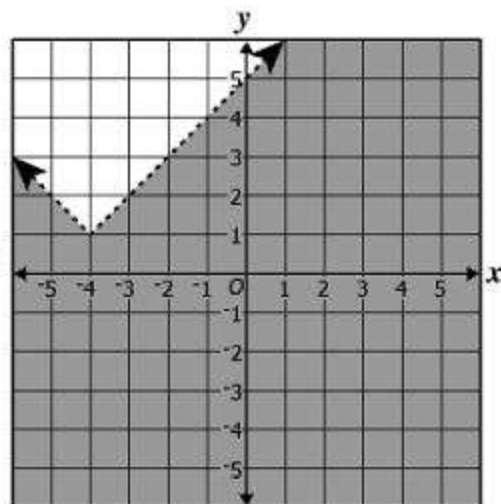
☐ C



☐ B



☐ D



What is a solution of $\sqrt{7-2x} + 5 = 8$?

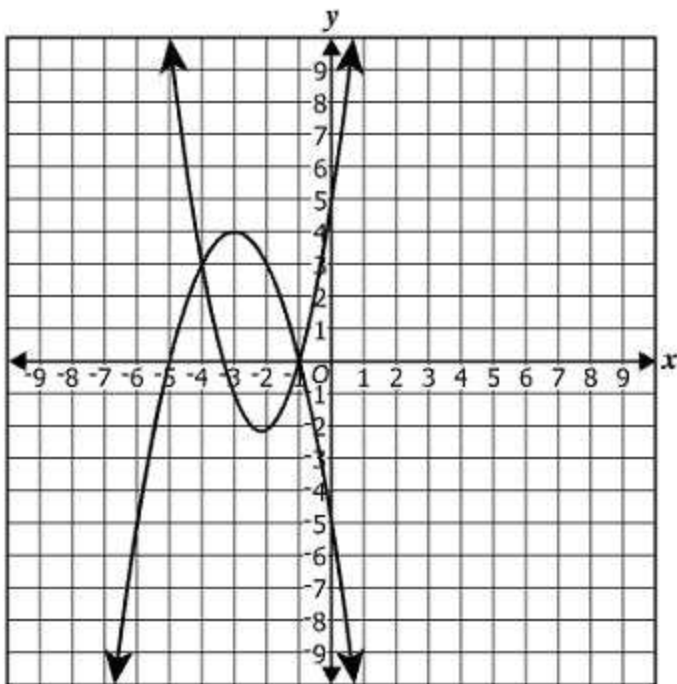
☐ **A** $x = -26$

☐ **B** $x = \frac{-19}{2}$

☐ **C** $x = \frac{-13}{2}$

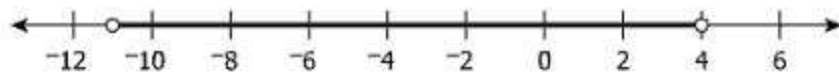
☐ **D** $x = -1$

Which is the apparent solution set of the system of equations graphed on the following grid?



- ☐ A $\{(0, -5), (0, 5)\}$
- ☐ B $\{(-3, 4), (-2, -2)\}$
- ☐ C $\{(-4, 3), (-1, 0)\}$
- ☐ D $\{(-5, 0), (-3\frac{1}{3}, 0), (-1, 0)\}$

This graph best represents the solution to which inequality?



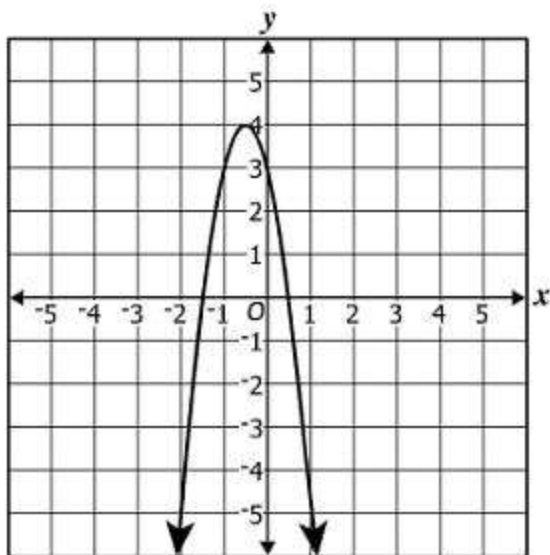
- ☐ A $|x - 11| > 4$
- ☐ B $|x - 11| < 4$
- ☐ C $|2x + 7| > 15$
- ☐ D $|2x + 7| < 15$

Directions: Type your answer in the box.

What value of x makes $\sqrt[3]{2x - 5} = 3$ true?

$x =$

What are the apparent roots of the equation graphed on the coordinate grid?



- ☐ A $\{0, 3\}$
- ☐ B $\left\{-\frac{1}{2}, 4\right\}$
- ☐ C $\left\{-\frac{3}{2}, \frac{1}{2}\right\}$
- ☐ D $\{-2, 1\}$

If no denominator is equal to zero, what is the solution set for the following equation?

$$\frac{3x - 4}{x^2} = \frac{3}{2x}$$

- ☐ A $\left\{\frac{8}{3}\right\}$
- ☐ B $\left\{\frac{8}{9}\right\}$
- ☐ C $\left\{-\frac{2}{3}, 2\right\}$
- ☐ D $\left\{-\frac{2}{3}, \frac{2}{3}\right\}$

What is the solution set for the following system of equations?

$$\begin{cases} y = 4x + 2 \\ y = x^2 + x - 8 \end{cases}$$

- ☐ A $\{(-5, -18), (2, 10)\}$
- ☐ B $\{(-1, -2), (6, 26)\}$
- ☐ C $\{(-6, -22), (1, 6)\}$
- ☐ D $\{(-2, -6), (5, 22)\}$

Directions: Click on all the correct answers.

Identify all the points where the graph of $h(x) = (x + 1)(x^2 + 8x + 16)$ intersects the x -axis.

☐ $(-4, 0)$	☐ $(1, 0)$
☐ $(-2, 0)$	☐ $(4, 0)$
☐ $(-1, 0)$	☐ $(16, 0)$

The function $f(x) = (1 - x)^2 - 4$ is decreasing throughout the interval —

- ☐ A $-4 < x < \infty$
- ☐ B $-\infty < x < 1$
- ☐ C $-1 < x < 3$
- ☐ D $-\infty < x < \infty$

Given: $f(x) = 4x^4 - 15$ and $g(x) = 2x + 11$

What is the value of $g(f(x))$?

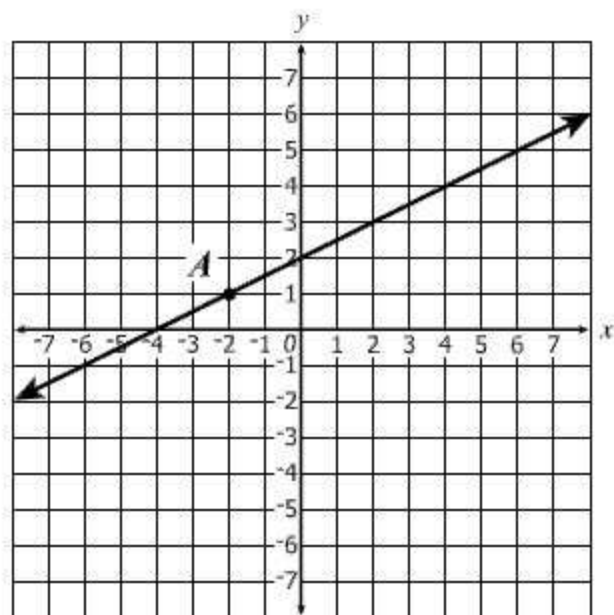
- ☐ A $8x^5 + 44x^4 - 30x - 165$
- ☐ B $8x^5 - 165$
- ☐ C $8x^4 - 4$
- ☐ D $8x^4 - 19$

A normally distributed data set has a mean of 0 and a standard deviation of 0.5 . Which is closest to the percent of values between -1 and 1 ?

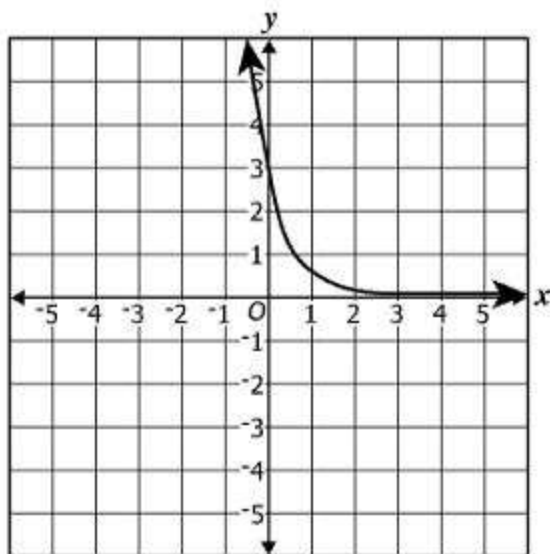
- ☐ A 34%
- ☐ B 50%
- ☐ C 68%
- ☐ D 95%

Directions: Click on the grid to plot the correct point.

Point A lies on the graph of $f(x) = \frac{1}{2}x + 2$. Locate the image of Point A that lies on the graph of $f^{-1}(x)$.



Which equation best represents this graph?



- ☐ A $f(x) = 3\left(\frac{1}{5}\right)^x$
- ☐ B $f(x) = 3\sqrt{5x}$
- ☐ C $f(x) = \frac{1}{3}\log(5x)$
- ☐ D $f(x) = \frac{1}{3}(5)^x$

If $f(x) = x^2 + 3x$ and $g(x) = 2x^2$, what is $g(f(-1))$?

☐ A -4

☐ B 0

☐ C 8

☐ D 10

The volume of a cone (V) varies jointly with its height (h) and the square of its radius (r). If k is the constant of proportionality, which of the following equations represents the correct relationship between volume, radius, and height?

☐ A $V = k(rh)^2$

☐ B $V = \frac{kr^2}{h}$

☐ C $V = \frac{k}{r^2h}$

☐ D $V = kr^2h$

What is the equation of the horizontal asymptote of the graph of the following equation?

$$f(x) = 4^{(x+1)} - 10$$

- ☐ A $y = 4$
- ☐ B $y = 0$
- ☐ C $y = -1$
- ☐ D $y = -10$

As x approaches negative infinity, which of the following describes the end behavior of $f(x) = -x^7 + bx^3 + c$?

- ☐ A $f(x)$ approaches c
- ☐ B $f(x)$ approaches 0
- ☐ C $f(x)$ approaches positive infinity
- ☐ D $f(x)$ approaches negative infinity

Jessica paid \$23,000 for her car and kept a record of its value.

Number of Years (x)	Value (in dollars) (y)
0	23,000
1	20,000
2	16,000
3	14,000
4	12,000
5	10,000

Assuming the relationship is exponential, which equation best models the curve of best fit for the data?

- ☐ A $y = 21,000(1.20)^x$
- ☐ B $y = 22,300(2.60)^x$
- ☐ C $y = 23,100(0.85)^x$
- ☐ D $y = 23,500(0.70)^x$

What is the sum of the infinite geometric series $9 - 6 + 4 - \frac{8}{3} + \dots$?

☐ A $\frac{29}{3}$

☐ B $\frac{25}{3}$

☐ C $\frac{27}{5}$

☐ D $\frac{18}{5}$

Which number is a zero of $f(x) = 7x^2 + 16x - 48$?

☐ A 12

☐ B 4

☐ C $\frac{12}{7}$

☐ D $\frac{4}{7}$

Which function is the inverse of $g(x) = x^3 + 11$?

☐ A $g^{-1}(x) = \sqrt[3]{x - 11}$

☐ B $g^{-1}(x) = \sqrt[3]{x + 11}$

☐ C $g^{-1}(x) = x - \sqrt[3]{11}$

☐ D $g^{-1}(x) = x + \sqrt[3]{11}$

What is the domain of $g(x) = \log(x - 1)$?

- ☐ A $\{x \mid x > 10\}$
- ☐ B $\{x \mid x > 9\}$
- ☐ C $\{x \mid x > 1\}$
- ☐ D $\{x \mid x > 0\}$

A scientist obtained a sample that contained 80 grams of radioactive Barium-122 that decays exponentially over time. The amount of Barium-122 that remained in the sample at observed times is shown in the table.

Radioactive Decay of Barium-122

Time (minutes)	Mass of Remaining Barium-122 (grams)
0	80.0
1	56.6
2	40.0
3	28.3
4	20.0

If the radioactive decay continues at the same rate, which is closest to the amount of the sample of Barium-122 remaining at 5 minutes?

- ☐ A 8.3 grams
- ☐ B 10.0 grams
- ☐ C 11.7 grams
- ☐ D 14.1 grams

What is the sum of the first 20 terms of the arithmetic sequence shown?

$$\frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \frac{5}{3}, \dots$$

- ☐ A 5
- ☐ B 20
- ☐ C 70
- ☐ D 140

Directions: Type your answer in the box.

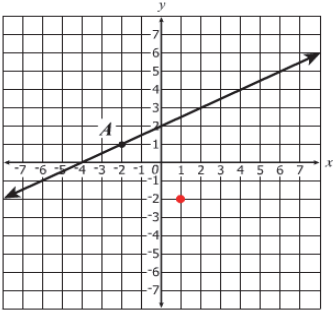
What is the number of possible permutations of 8 objects taken 3 at a time?

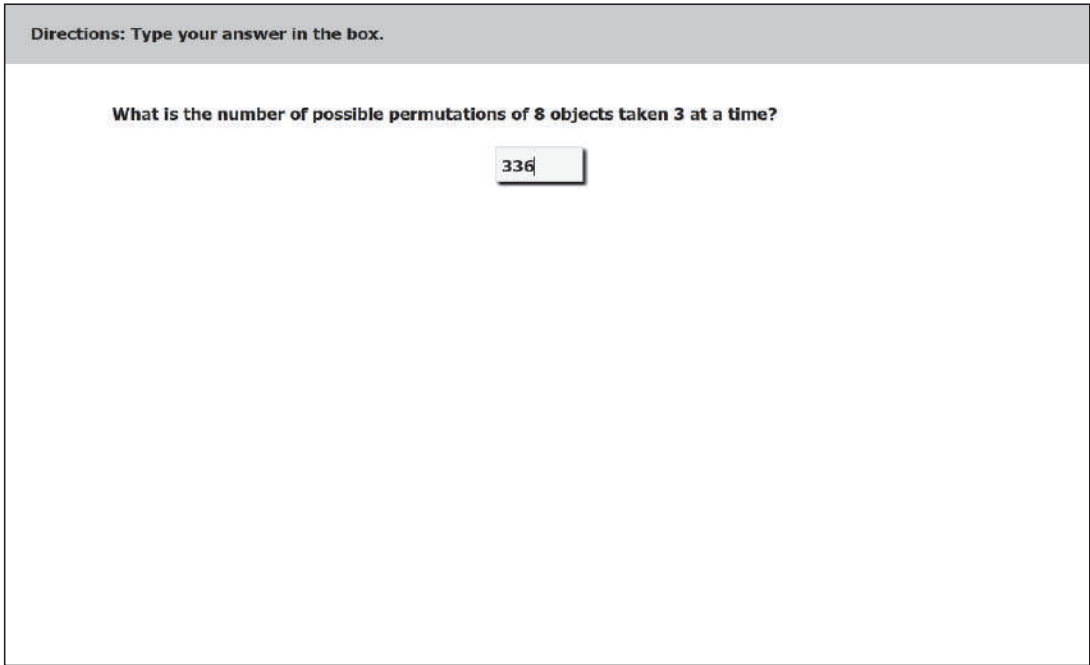
Algebra II
Released Test Item Set Spring 2015
Answer Key

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
1	MC	C	001	Expressions and Operations
2	MC	A	001	Expressions and Operations
3	MC	A	001	Expressions and Operations
4	MC	A	001	Expressions and Operations
5	MC	D	001	Expressions and Operations
6	MC	B	001	Expressions and Operations
7	MC	C	001	Expressions and Operations
8	MC	C	001	Expressions and Operations
9	MC	A	001	Expressions and Operations
10	MC	B	001	Expressions and Operations
11	MC	A	002	Equations and Inequalities
12	MC	B	002	Equations and Inequalities
13	MC	B	002	Equations and Inequalities
14	MC	D	002	Equations and Inequalities
15	MC	C	002	Equations and Inequalities
16	MC	D	002	Equations and Inequalities

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
17	TEI	Typed response: 16 (and all equivalent answers) Directions: Type your answer in the box. What value of x makes $\sqrt[3]{2x-5} = 3$ true? $x =$ 16	002	Equations and Inequalities
18	MC	C	002	Equations and Inequalities
19	MC	A	002	Equations and Inequalities
20	MC	D	002	Equations and Inequalities

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description						
21	TEI	(−4,0) (first row, left column) and (−1,0) (last row, left column) Both of these answers, and only these answers, must be selected. Directions: Click on all the correct answers. Identify all the points where the graph of $h(x) = (x + 1)(x^2 + 8x + 16)$ intersects the x-axis. <table><tr><td>☒ (−4,0)</td><td>☐ (1, 0)</td></tr><tr><td>☐ (−2,0)</td><td>☐ (4, 0)</td></tr><tr><td>☒ (−1,0)</td><td>☐ (16, 0)</td></tr></table>	☒ (−4,0)	☐ (1, 0)	☐ (−2,0)	☐ (4, 0)	☒ (−1,0)	☐ (16, 0)	003	Functions and Statistics
☒ (−4,0)	☐ (1, 0)									
☐ (−2,0)	☐ (4, 0)									
☒ (−1,0)	☐ (16, 0)									
22	MC	B	003	Functions and Statistics						
23	MC	D	003	Functions and Statistics						
24	MC	D	003	Functions and Statistics						

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
25	TEI	A point must be plotted on the coordinate plane at (1,−2). This point is the only correct answer. Directions: Click on the grid to plot the correct point. Point A lies on the graph of $f(x) = \frac{1}{2}x + 2$. Locate the image of Point A that lies on the graph of $f^{-1}(x)$. 	003	Functions and Statistics
26	MC	A	003	Functions and Statistics
27	MC	C	003	Functions and Statistics
28	MC	D	003	Functions and Statistics
29	MC	D	003	Functions and Statistics
30	MC	C	003	Functions and Statistics
31	MC	C	003	Functions and Statistics
32	MC	C	003	Functions and Statistics

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
33	MC	C	003	Functions and Statistics
34	MC	A	003	Functions and Statistics
35	MC	C	003	Functions and Statistics
36	MC	D	003	Functions and Statistics
37	MC	C	003	Functions and Statistics
38	TEI	Typed response: 336 	003	Functions and Statistics

Algebra II Practice Test

Objective: 1.1a

1. Which is equivalent to $49^{\frac{3}{2}}$?

- A 21
- B 98
- C 294
- D 343

2. Which expression is another way to write $\sqrt[3]{125x^4}$?

- A $5x^{\frac{4}{3}}$
- B $5x^{\frac{4}{3}}$
- C $25x^{\frac{4}{3}}$
- D $25x^{\frac{4}{3}}$

3. If x and y are real numbers, what is the simplified radical form of $(x^2y^5)^{\frac{1}{5}}$?

- A $y\sqrt[5]{x^2}$
- B $y\sqrt{x^5}$
- C $|y|\sqrt[5]{x^2}$
- D $|y|\sqrt{x^5}$

Objective 1.1b

4. What is the simplified expression of $\sqrt{\frac{36x^8}{4x^6}}$?

- A $3x$
- B $9x$
- C $3x^2$
- D $9x^2$

5. What is the simplified form of $(2\sqrt{5}+3)(\sqrt{5}-1)$?

- A $\sqrt{5}-3$
- B $\sqrt{5}+7$
- C $2\sqrt{5}-3$
- D $2\sqrt{5}+7$

6. What is the sum of $\frac{1}{3\sqrt{25}}$ and $\frac{1}{2\sqrt[3]{27}}$?

- A $\frac{2}{21}$
- B $\frac{7}{30}$
- C $\frac{2}{33}$
- D $\frac{11}{90}$

7. The area of a square is $2\sqrt{2}+3$. What is the length of a side of the square?

- A $\sqrt{2}-1$
- B $\sqrt{2}+1$
- C $2\sqrt{2}-1$
- D $2\sqrt{2}+1$

Objective 1.2a

8. Which expression represents the quotient? $\frac{8x^6z^4 + 4x^4z^2}{4x^2z}$

- A $2x^4z^3 + x^2z$
- B $2x^3z^4 + x^2z^2$
- C $4x^4z^3 + 3x^2z$
- D $4x^3z^4 + 3x^2z^2$

9. Which expression represents the quotient? $\frac{4x^2y}{8xy^2} \div \frac{12xy^2}{8x^6y^3}$

- A $\frac{x^5}{3}$
- B $\frac{3}{x^5}$
- C $\frac{x^6}{3}$
- D $\frac{3}{x^6}$

10. Which expression represents the quotient? $(y^2 - 4y - 32) \div (y + 4)$

- A $y-8$
- B $y+8$
- C $y-4$
- D $y+4$

11. A rectangular prism has a volume of $8x^3 + 14x^2 + x - 2$ and a height of $2x + 1$. Which expression represents the area of the base of the prism?

A $4x^2 + 5x - 2$
B $4x^2 + 5x + 2$
C $4x^2 + 9x + 4$
D $4x^2 + 9x + 5$

objective 1.2b

12. What is the completely simplified equivalent of $\frac{x^2 + x - 12}{x^2 - 6x + 9}$?

A $\frac{x-3}{x+4}$
B $\frac{x+4}{x-3}$
C $2x^2 + 5x - 3$
D $2x^2 + 7x + 21$

13. Which expression represents the result of this subtraction $\frac{3x-1}{x+2} - \frac{x-2}{x-1}$?

A $\frac{2x+1}{3}$
B $\frac{2x+1}{x^2 + x - 2}$
C $\frac{3x^2 - 4x + 5}{3}$
D $\frac{2x^2 - 4x + 5}{x^2 + x - 2}$

14. What is the simplified equivalent of $2 - x - \frac{1}{3-x}$?

A $\frac{1}{3-2x}$
B $\frac{x^2 - x + 3}{3-x}$
C $\frac{x^2 - 5x + 5}{3-x}$
D $\frac{x^2 - 5x + 7}{3-x}$

objective 1.3b

15. Which expression is equivalent to $(4i)^3$?

- A $-12i$
- B $12i$
- C $-64i$
- D $64i$

16. A circuit has a current of $(8 + 7i)$ amps, and another circuit has a current of $(5 - 3i)$ amps. What is the difference between the currents of the two circuits?

- A $(3 - 4i)$ amps
- B $(3 + 4i)$ amps
- C $(3 - 10i)$ amps
- D $(3 + 10i)$ amps

17. Which expression is equivalent to $\sqrt{-6}(\sqrt{-4} - \sqrt{3})$?

- A $2\sqrt{6} + 3\sqrt{2}$
- B $-24 - 6i\sqrt{3}$
- C $2\sqrt{6} - 3i\sqrt{2}$
- D $-2\sqrt{6} - 3i\sqrt{2}$

18. What is the product of $(2 + 3i)$ and $(5 - 4i)$?

- A $-2 - 23i$
- B $-2 + 7i$
- C $22 - 23i$
- D $22 + 7i$

19. What is the completely simplified equivalent of $\frac{2}{5+i}$?

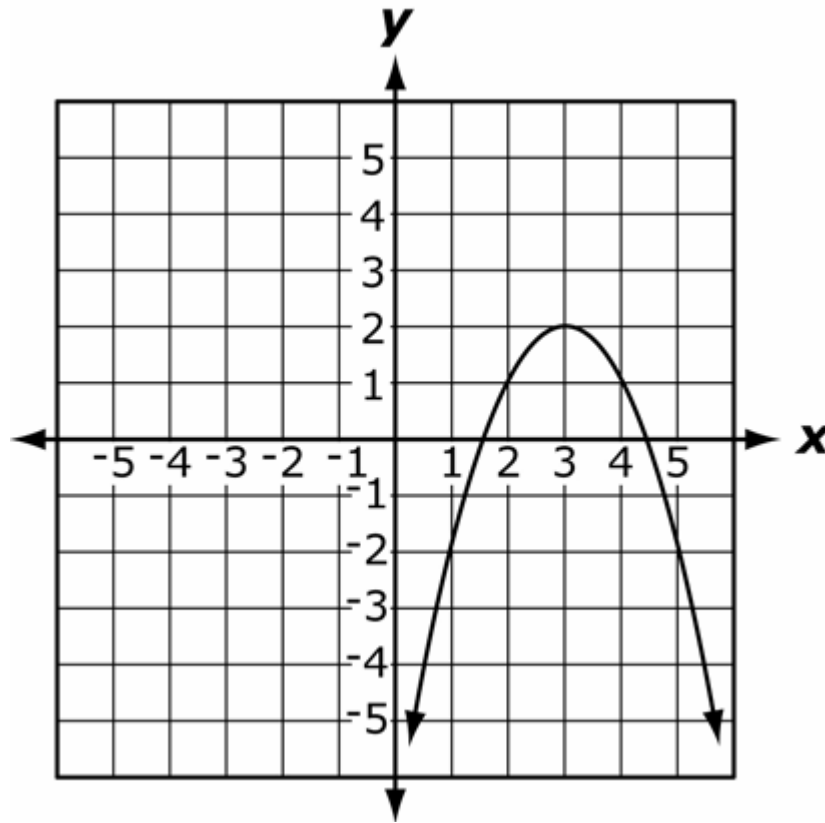
- A $\frac{5-i}{12}$
- B $\frac{5+i}{12}$
- C $\frac{5-i}{13}$
- D $\frac{5+i}{13}$

objective 2.1a

20. What is the parent graph of the following function and what transformations have taken place on it: $y = (x-3)^2$?

- A The parent graph is $y = x^2$, which is shifted 3 units up.
- B The parent graph is $y = x^2$, which is shifted 3 units down.
- C The parent graph is $y = x^2$, which is shifted 3 units to the left.
- D The parent graph is $y = x^2$, which is shifted 3 units to the right.

21. What is the parent function of this graph?



- A $f(x) = x^2$
- B $f(x) = x^4$
- C $f(x) = -x^2$
- D $f(x) = -x^4$

objective 2.1b

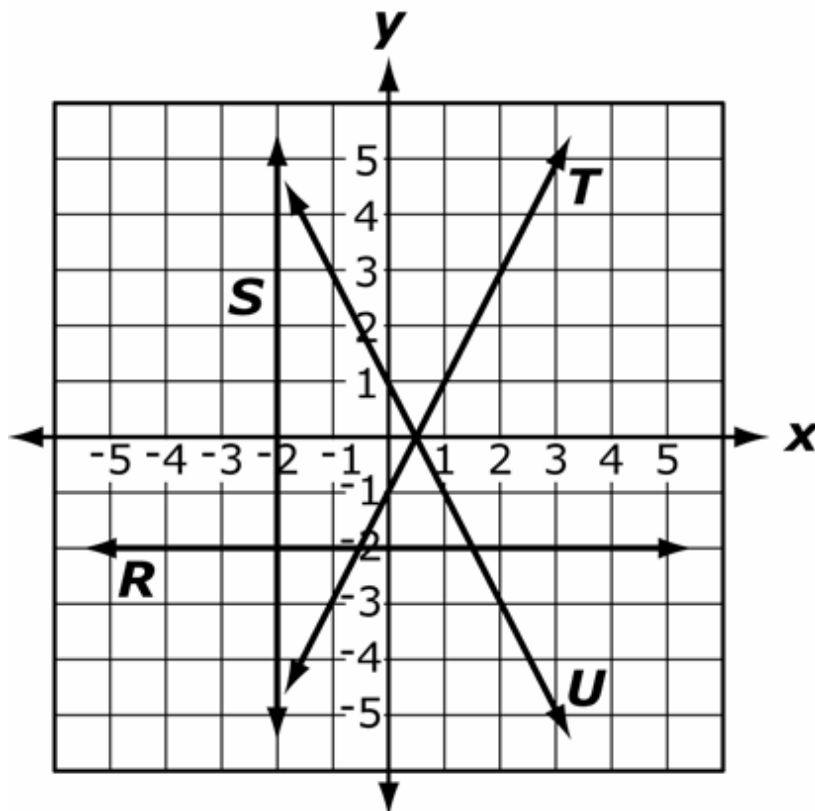
22. If $f(x) = 3x^2 - 2$ and $g(x) = 4x + 2$, what is the value of $(f + g)(-1)$?

- A -7
- B -1
- C 1
- D 7

23. If $f(x) = x^2 - 1$ and $g(x) = x - 1$, what is the value of $\left(\frac{f}{g}\right)(x)$?

- A $x - 1$
- B $x + 1$
- C $\frac{1}{x - 1}$
- D $\frac{1}{x + 1}$

24.



If $f(x) = x - \frac{1}{2}$ and $g(x) = -2$, which graph corresponds to the function of $(fg)(x)$?

- A line R
- B line S
- C line T
- D line U

Objective 2.1c

25. If $f(x) = 2x + 7$ and $g(x) = 3x^2 - 1$, what expression represents $(f(g(x)))$?

- A $6x^2 + 5$
- B $6x^2 + 12$
- C $3x^2 - 2x - 8$
- D $3x^2 + 2x + 6$

26. If $(f \circ g)(x) = 2x - 1$, how might $f(x)$ and $g(x)$ be defined?

- A $f(x) = (x - 1)$ and $g(x) = (2x - 1)$
- B $f(x) = (x - 1)$ and $g(x) = (2x + 1)$
- C $f(x) = (2x - 1)$ and $g(x) = (x - 1)$
- D $f(x) = (2x + 1)$ and $g(x) = (x - 1)$

27. Which statement is true for the function $f(x) = \frac{1}{x+4}$?

- A 4 is not in the range of the function.
- B 4 is not in the domain of the function.
- C -4 is not in the range of the function.
- D -4 is not in the domain of the function.

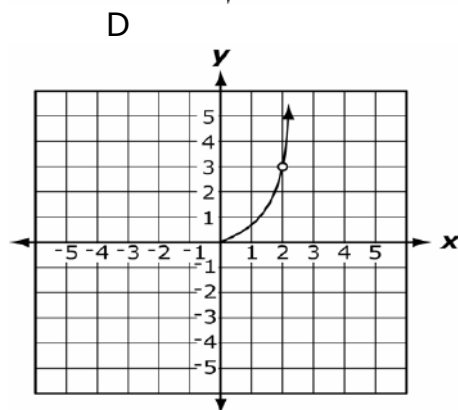
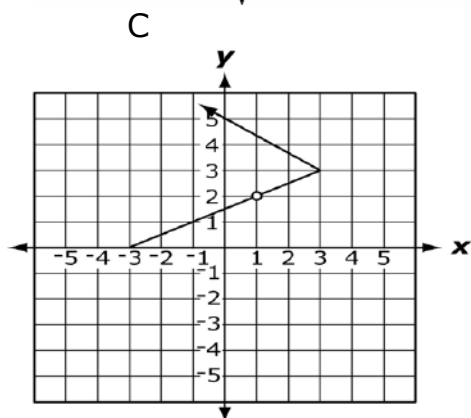
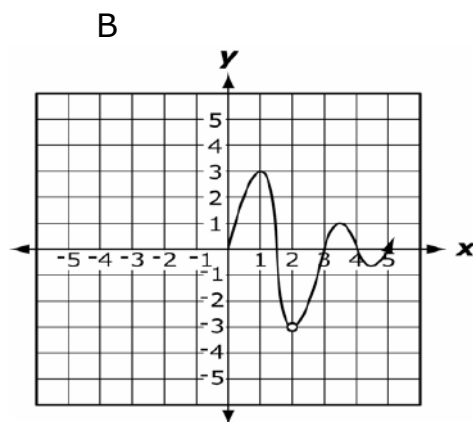
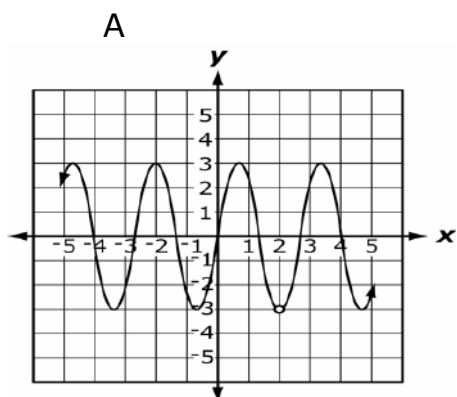
28. What is the domain of the function $f(x) = \frac{x+5}{x^2+2x-8}$?

- A $\{x: x \neq 0\}$
- B $\{x: x \neq -5\}$
- C $\{x: x \neq -2, 4\}$
- D $\{x: x \neq 2, -4\}$

29. Which intervals correctly define the domain of $f(x) = \frac{1}{x+4} - 2$?

- A $(-\infty, 4)$ and $(4, \infty)$
- B $(-\infty, -4)$ and $(4, \infty)$
- C $(-\infty, -4)$ and $(-4, \infty)$
- D $(-\infty, -4)$ and $(-2, \infty)$

30. Domain: $\{x|x \geq 0, x \neq 2\}$ Range: $\{y|-3 < y \leq 3\}$ Which graph corresponds to the given constraints?



31. Which function has the fewest domain restrictions for real numbers?

A $f(x) = \frac{1}{x-1}$

B $f(x) = \frac{1}{x+1}$

C $f(x) = \frac{1}{x^2-1}$

D $f(x) = \frac{1}{x^2+1}$

Objective 2.1e

32. What is the inverse of $f(x) = x+1$?

A $f^{-1}(x) = -x-1$

B $f^{-1}(x) = x-1$

C $f^{-1}(x) = \frac{-1}{1-x}$

D $f^{-1}(x) = \frac{1}{1+x}$

33. What is the inverse of the function $f(x) = (x+4)^2$?

A $f^{-1}(x) = \sqrt{x} - 4$

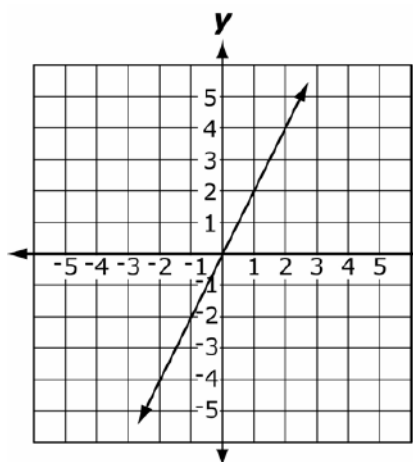
B $f^{-1}(x) = \frac{1}{(x+4)^2}$

C $f^{-1}(x) = \pm\sqrt{x} - 4$

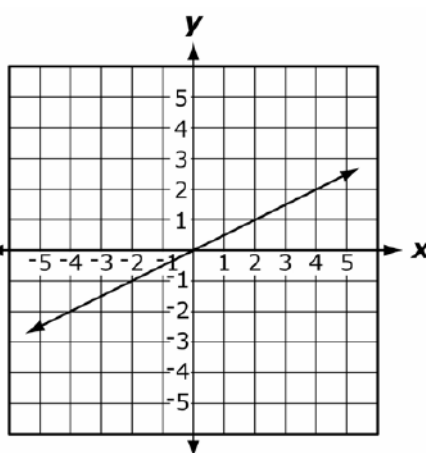
D $f^{-1}(x) = (x-4)^2$

34. Which graph represents the inverse of $f(x) = 2x$?

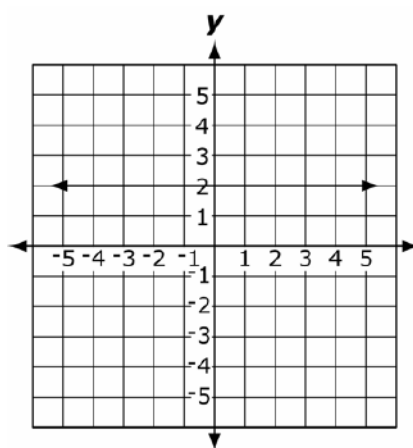
A



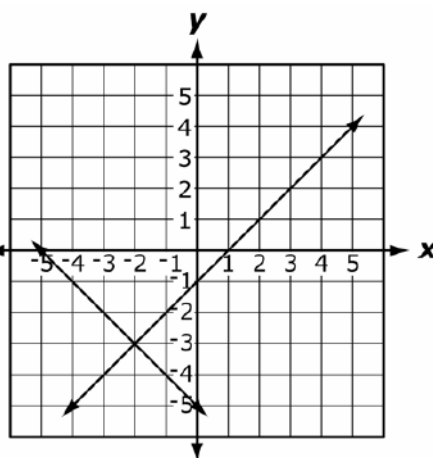
B



C



D



35. Which statement about graphs and their inverse is true?

A They are symmetric about $y = x$.

B They are symmetric about the origin.

C They are symmetric about the x-axis.

D They are symmetric about the y-axis.

Objective 2.2a

36. Profits, P , are equal to sales, S , minus expenses, E . If expenses are equal to travel, T , plus materials, M , which system of equations models this situation?

A	$P = S - E$ $E = T + M$	B	$P = S + E$ $E = T + M$
C	$P = S - E$ $E = T - M$	D	$P = S + E$ $E = T - M$

37. Tyrone wants to spend at most \$10,000 on two televisions, R and S . Each television must cost at least \$3,000, and television R must cost at least twice as much as television S . Which system of inequalities models the amount of money spent on each television?

A	$R + S \geq 10,000$ $R \geq 2S$ $R \geq 3,000$ $S \geq 3,000$	B	$R + S \leq 10,000$ $S \geq 2R$ $R \geq 3,000$ $S \geq 3,000$
C	$R + S \leq 10,000$ $R \geq 2S$ $R \geq 3,000$ $S \geq 3,000$	D	$R + S \geq 10,000$ $S \geq 2R$ $R \geq 3,000$ $S \geq 3,000$

38. Meredith invests \$50,000 in her new business. It costs the company \$10 to produce each unit, which is sold for \$15. Let C represent the cost and R represent the revenue for x units. Which statement is true about the graphs of the equations $C = 50,000 + 10x$ and $R = 15x$?

- A Both slopes are positive.
- B Both slopes are negative.
- C One slope is positive, and the other is zero.
- D one slope is negative, and the other is positive.

Objective 2.2b

39. Which quadrants contain the solutions to this system of inequalities?

$$\begin{cases} y - 2x \leq -3 \\ 3y + x \geq -4 \end{cases}$$

- A quadrants I and IV
- B quadrants II and III
- C quadrants III and IV
- D quadrants II, III, and IV

40. What is the solution to this system of equations? $\begin{cases} 3x - y + 5 = 0 \\ 2x + 3y - 4 = 0 \end{cases}$

A $x = -1, y = -2$

B $x = -1, y = 2$

C $x = 2, y = -1$

D $x = 2, y = 1$

41. The corners of a triangle are $(2,1)$, $(4,4)$, and $(6,2)$. Which system of inequalities describes the interior of the triangle?

A $\begin{cases} 4y > x + 2 \\ 3y < 2x - 1 \\ y < 8 - x \end{cases}$

B $\begin{cases} 4y > x + 2 \\ 2y < 3x - 4 \\ y < 8 - x \end{cases}$

C $\begin{cases} 2y > x \\ 2y < 3x - 4 \\ y < 8 - x \end{cases}$

D $\begin{cases} 2y > x \\ 3y < 2x - 1 \\ y < 8 - x \end{cases}$

Objective 2.2c

42. What is the solution set of this system of equations?

$$\begin{cases} x^2 + y - 1 = 0 \\ x - y + 1 = 0 \end{cases}$$

A $\{(-1,-1), (-1,0)\}$

B $\{(-1,0), (-1,1)\}$

C $\{(-1,0), (0,1)\}$

D $\{(1,0), (1,1)\}$

43. What is the solution set of this system of equations?

$$\begin{cases} y - x = 3 \\ x^2 - 7y + 31 = 0 \end{cases}$$

A $\{(2,5), (5,2)\}$

B $\{(2,5), (5,8)\}$

C $\{(5,8), (8,5)\}$

D $\{(8,5), (8,8)\}$

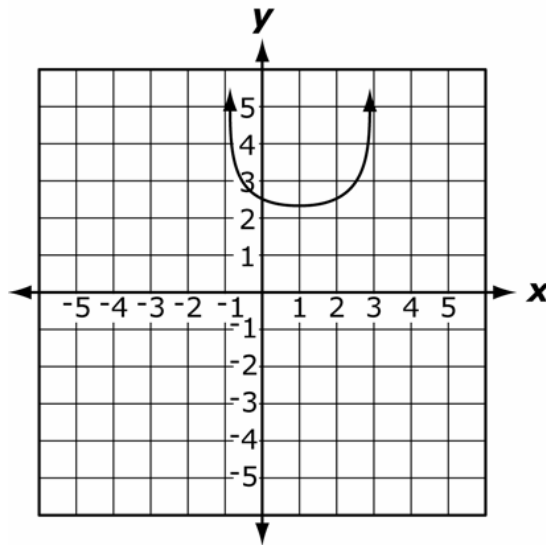
44. What is the solution set of this system of equations?

$$\begin{cases} x^2 - y = -3 \\ 2x^2 - y = -2 \end{cases}$$

- A $\{(-1, -4), (-1, 4)\}$
- B $\{(-1, -4), (1, 4)\}$
- C $\{(-1, 4), (1, -4)\}$
- D $\{(-1, 4), (1, 4)\}$

Objective 2.3a

45. How many real roots does the function given by the graph have?



- A 0 real roots
- B 1 real root
- C 2 real roots
- D 4 real roots

46. What number is added to both sides of the equation $x^2 - 8x + 3 = 0$ to solve it by completing the square?

- A -16
- B 16
- C -64
- D 64

47. What is the solution of $x^2 + 5x - 3 = 0$?

- A $\frac{-5 \pm \sqrt{13}}{2}$
- B $\frac{-5 \pm \sqrt{37}}{2}$
- C $\frac{5 \pm \sqrt{13}}{2}$
- D $\frac{5 \pm \sqrt{37}}{2}$

Objective 2.3b

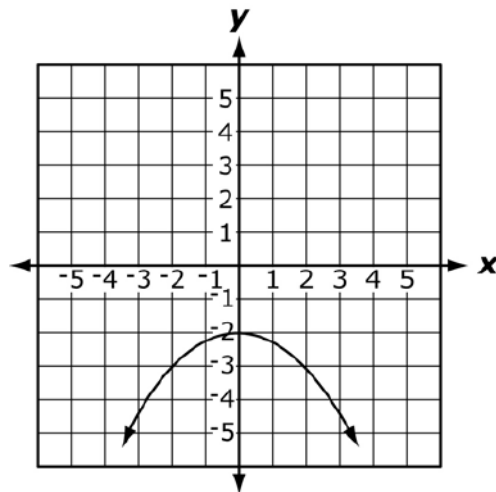
48. What is the y-intercept of $f(x) = 3x^2 - 2x + 1$?

- A (0, -1)
- B (0, 1)
- C (-1, 0)
- D (1, 0)

49. What are the coordinates at the minimum point of $f(x) = x^2 - 4x + 3$?

- A (-1, -2)
- B (-1, 2)
- C (2, -1)
- D (2, 1)

50. Which function represents this graph?



- A $f(x) = \frac{-1}{4}x^2 - 2$
- B $f(x) = \frac{1}{4}x^2 - 2$
- C $f(x) = -4x^2 - 2$
- D $f(x) = 4x^2 - 2$

51. Which statement best describes these two functions?

$$f(x) = x^2 - x + 6$$

$$g(x) = -3x^2 + 3x + 5$$

- A They have no common points.
- B They have the same x-intercepts.
- C The maximum of $f(x)$ is the same as the minimum of $g(x)$.
- D The maximum of $g(x)$ is the same as the minimum of $f(x)$.

52. Which statement best describes these two functions?

$$f(x) = x^2 - x + 4$$

$$g(x) = -3x^2 + 3x + 7$$

- A The maximum of $f(x)$ is less than the minimum of $g(x)$.
- B The minimum of $f(x)$ is less than the maximum of $g(x)$.
- C The maximum of $f(x)$ is greater than the minimum of $g(x)$.
- D The minimum of $f(x)$ is greater than the maximum of $g(x)$.

Objective 2.3c

53. The length of a rectangular swimming pool is 20 feet greater than the width. The surface area of the pool is 1,500 square feet. What are the length and width of the pool?

- A length = 20 ft, width = 20 ft
- B length = 50 ft, width = 30 ft
- C length = 60 ft, width = 40 ft
- D length = 150 ft, width = 10 ft

54. The profit, P , (in dollars) for Ace Car Rental is given by $P = 100x - 0.1x^2$, where x is the number of cars rented. How many cars have to be rented for the company to maximize profits?

- A 500 cars
- B 1,000 cars
- C 12,500 cars
- D 25,000 cars

55. The revenue, R , at a bowling alley is given by the equation $R = \frac{-1}{800}(x^2 - 2,400x)$, where x is the number of frames bowled.

What is the maximum amount of revenue the bowling alley can generate?

- A \$800
- B \$1,200
- C \$1,800
- D \$2,400

56. Which best describes the graph of $\frac{x^2}{50} + \frac{y^2}{25} = 1$?

- A circle
- B ellipse
- C parabola
- D hyperbola

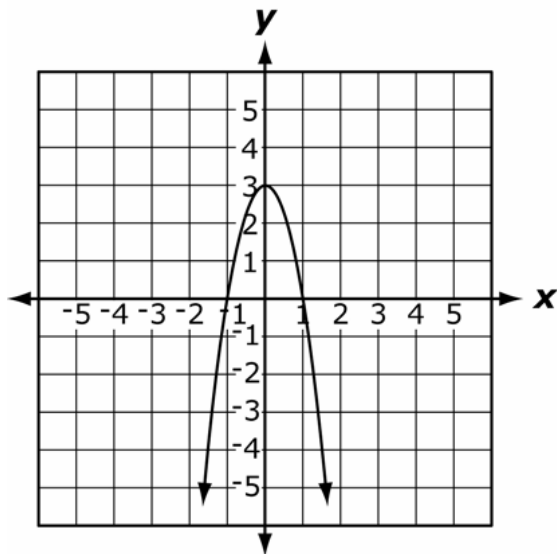
57. What is the equation of a circle with center $(-4, 2)$ and diameter 6?

- A $(x-4)^2 + (y+2)^2 = 6$
- B $(x+4)^2 + (y-2)^2 = 6$
- C $(x-4)^2 + (y+2)^2 = 9$
- D $(x+4)^2 + (y-2)^2 = 9$

58. Which statement describes the equation $y = -x^2 + 6x - 8$?

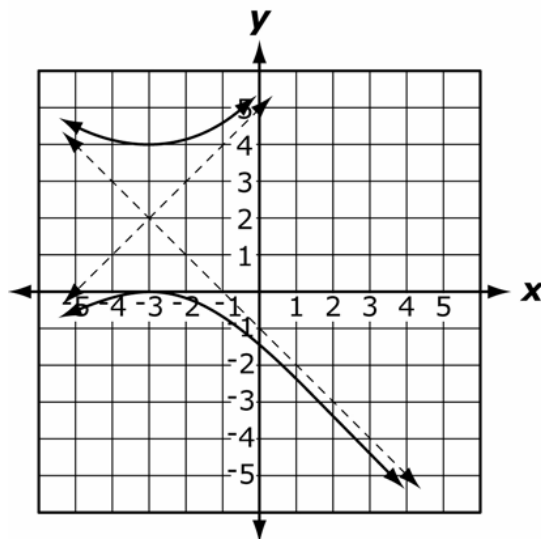
- A It is a vertical parabola.
- B It is a vertical hyperbola.
- C It is a horizontal parabola.
- D It is a horizontal hyperbola.

59. What is the equation of the given parabola?



- A $y = -x^2 + 3$
- B $y = -3x^2 + 3$
- C $y = -x^2 - 2x + 3$
- D $y = 3x^2 - 6x + 3$

60. What is the equation of the graphed Hyperbola?



A $\frac{(x+3)^2}{4} - \frac{(y-2)^2}{4} = 1$

B $\frac{(y+2)^2}{4} - \frac{(x-3)^2}{4} = 1$

C $\frac{(y-2)^2}{2} - \frac{(x+3)^2}{2} = 1$

D $\frac{(y-2)^2}{4} - \frac{(x+3)^2}{4} = 1$

61. What is the vertex of the parabola $y = (x-1)^2 - 9$?

A (-1, -9)

B (1, -9)

C (-9, -1)

D (-9, 1)

62. What is the equation of the ellipse whose center is at the origin, major axis has length of 10 units along the x-axis, and minor axis has length of 6 units?

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

A $\frac{x^2}{25} + \frac{y^2}{9} = 1$

B $\frac{x^2}{9} + \frac{y^2}{25} = 1$

C $\frac{x^2}{20} + \frac{y^2}{12} = 1$

D $\frac{x^2}{100} + \frac{y^2}{36} = 1$

63. Which function is best represented by the data in this table?

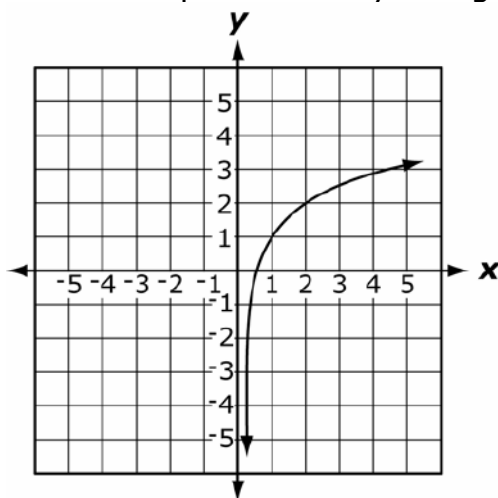
X	0	1	2	3	4
Y	1	3	9	27	81

- A $f(x) = x^3$
- B $f(x) = 3^x$
- C $f(x) = 3x$
- D $f(x) = 3x^2$

64. What are the horizontal asymptote and y-intercept for the graph of this function $f(x) = 2^{-x} + 7$?

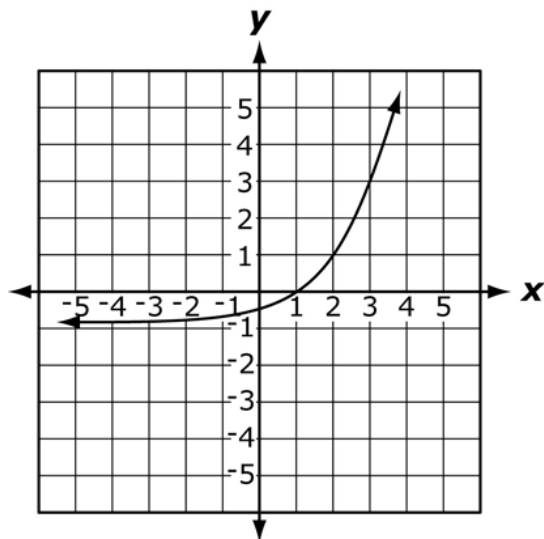
- A Asymptote: $y=7$, Intercept: $(0, 7)$
- B Asymptote: $y=-7$, Intercept: $(0, 7)$
- C Asymptote: $y=7$, Intercept: $(0, 8)$
- D Asymptote: $y=-7$, Intercept: $(0, 8)$

65. Which function is best represented by this graph?



- A $f(x) = \log_2 x + 1$
- B $f(x) = \log_2 x - 1$
- C $f(x) = \log_2(x+1)$
- D $f(x) = \log_2(x-1)$

66. Which function is best represented by this graph?



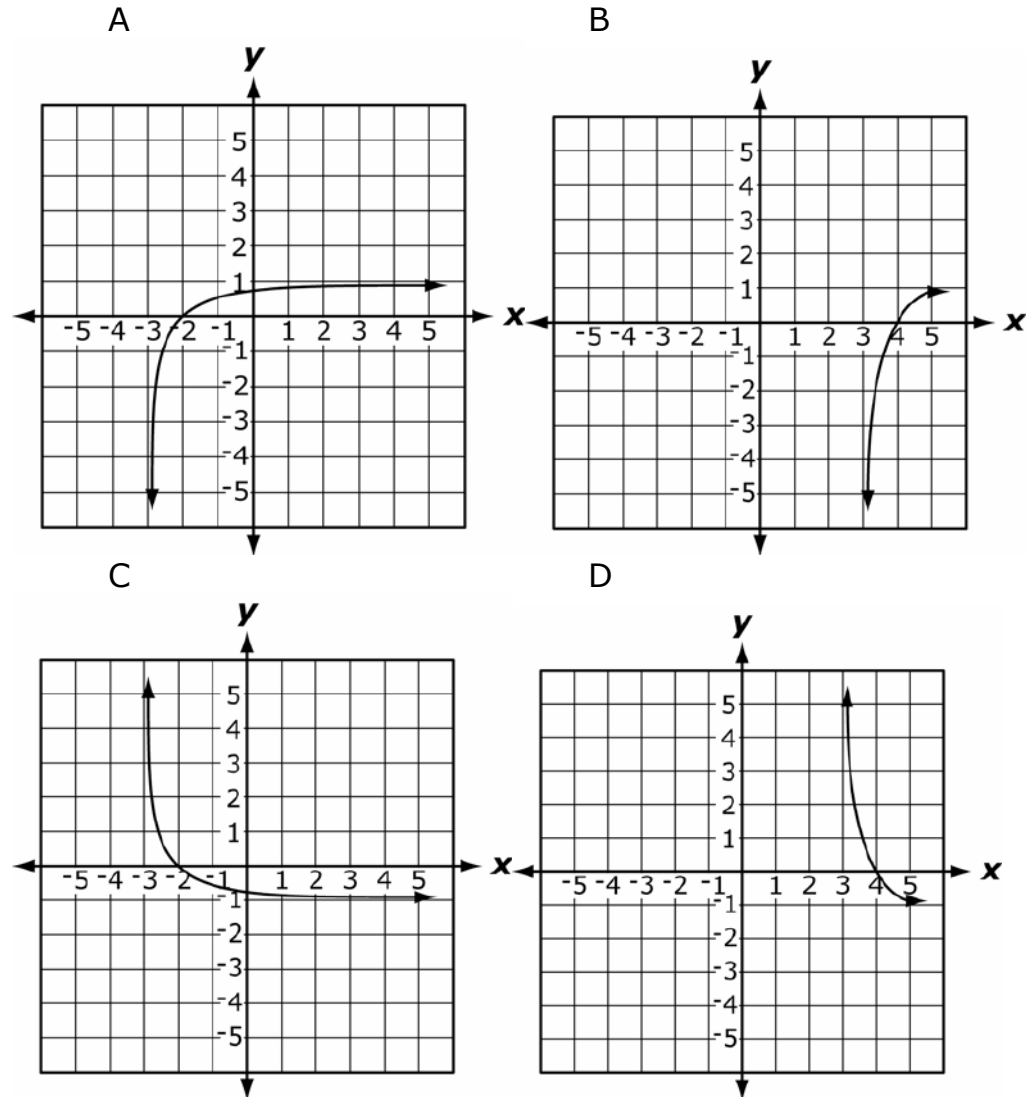
A $f(x) = 2^{x-1} - 1$

B $f(x) = 2^{x+1} - 1$

C $f(x) = 2^x - \frac{1}{2}$

D $f(x) = 2^{x-1}$

67. Which graph represents the function $f(x) = \log(x+3)$?



objective 2.5b

68. Which function is the inverse of $f(x) = \log x$?

- A $f(x) = e^x$
- B $f(x) = 2^x$
- C $f(x) = 10^x$
- D $f(x) = \frac{1}{\log x}$

69. If $3^{\log_3 7} = x$, what is the value of x ?

- A 7
- B 3^7
- C $\sqrt[3]{7}$
- D $\sqrt[7]{3}$

70. Which equation represents the solution for x in the formula $6^x = 21$?

A $x = \frac{\log 6}{\log 21}$

B $x = \frac{\log 21}{\log 6}$

C $x = \log 21 - \log 6$

D $x = \log 21 + \log 6$

71. What is the value of $\log \sqrt{10}$?

A 0

B $\frac{1}{2}$

C 1

D 10

72. If $\log_{2x} 80 = 2$, what is the value of x ?

A 20

B $2\sqrt{5}$

C $5\sqrt{2}$

D $2\sqrt{10}$

73. If $4\left(\log_3 \frac{1}{27}\right) = x$, what is the value of x ?

A $\frac{4}{3}$

B $-\frac{4}{3}$

C 12

D -12

Objective: 2.5c

74. If the loudness of fizz in a can of soda pop is represented

by $F = 4\log\left(\frac{x}{10^{-5}}\right)$, where x is represented by the intensity of sound, how

loud is the fizz if $x = 10^{-3}$?

A 4 decibels

B 8 decibels

C 16 decibels

D 32 decibels

75. The formula, $r = 2^{\frac{1}{x}} - 1$, gives the annual interest rate, r , required for your money to double in x years. If it takes 18 years for your money to double, what was the approximate annual interest rate?

- A 2%
- B 4%
- C 8%
- D 18%

76. The population, P , of prairie dogs increases according to the equation $P = 2,250e^{rt}$, where t is the number of years, and r is the rate of growth. Which equation solves for r ?

A $r = \frac{\ln\left(\frac{P}{2,250}\right)}{t}$

B $r = \frac{t}{\ln\left(\frac{P}{2,250}\right)}$

C $r = \frac{\ln\left(\frac{2,250}{P}\right)}{t}$

D $r = \frac{t}{\ln\left(\frac{2,250}{P}\right)}$

77. The mass of a radioactive sample is given by $M(t) = M_0 10^{-kt}$, where t is the time in years, M_0 is the initial mass, and k is a constant. If 400 grams of this material decays to 40 grams in 10 years, what is the value of k ?

- A 1
- B -1
- C 0.1
- D -0.1

Objective 2.6a

78. Which equation has -1 and 3 as solutions?

- A $x^2 - 2x - 3 = 0$
- B $x^2 - 2x + 3 = 0$
- C $x^2 + 2x - 3 = 0$
- D $x^2 + 2x + 3 = 0$

79. Which of these is a root of $f(x) = x^3 - 3x^2 - 4x + 12$?

- A -3
- B 3
- C 4
- D 12

80. Given that $(2x-1)$ and $(x+3)$ are factors of the polynomial, $2x^3 + 13x^2 + 17x - 12$, what is the third factor?

- A $x-4$
- B $x+4$
- C $3-x$
- D $3+x$

81. What is the solution set of $10x^2 - x - 3 = 0$?

- A $\left\{\frac{-1}{2}, \frac{3}{5}\right\}$
- B $\left\{\frac{-3}{5}, \frac{1}{2}\right\}$
- C $\left\{\frac{-3}{2}, \frac{1}{5}\right\}$
- D $\left\{\frac{-1}{5}, \frac{3}{2}\right\}$

82. A rectangular prism has a volume of 120 cubic inches. The length of the prism is 5 inches, the width is $(x-2)$ inches, and the height is $(x+3)$ inches. What are the width and height of the prism?

- A width: 3 in., height: 8 in.
- B width: 4 in., height: 6 in.
- C width: 6 in., height: 4 in.
- D width: 8 in., height: 3 in.

83. What is $(3x^5 - 15x^4 + 4x^3 + 11x^2 - 9x + 2)$ divided by $(x^2 - 5x + 2)$?

- A $(3x^3 - 2x + 1)$
- B $(3x^3 - 2x^2 + 7)$
- C $(3x^3 - 2x^2 + 7x + 26)$
- D $(3x^3 - 30x^2 + 160x - 849)$

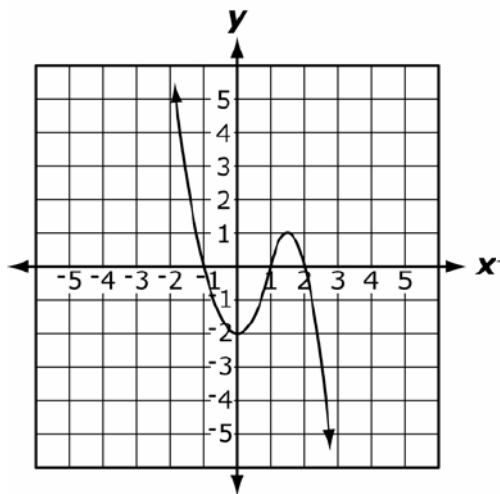
Objective 2.6b

84. In which direction does the graph of the parabola $x = -y^2$ open?

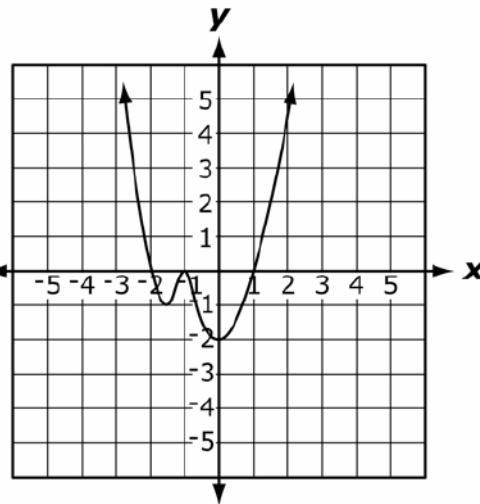
- A up
- B left
- C right
- D down

85. What is the graph of the polynomial $y = x^3 + 2x^2 - x - 2$?

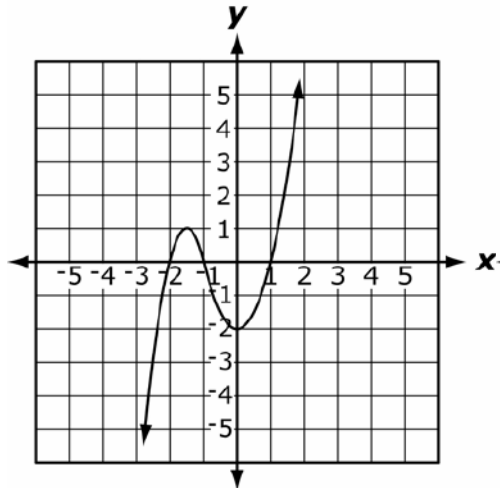
A



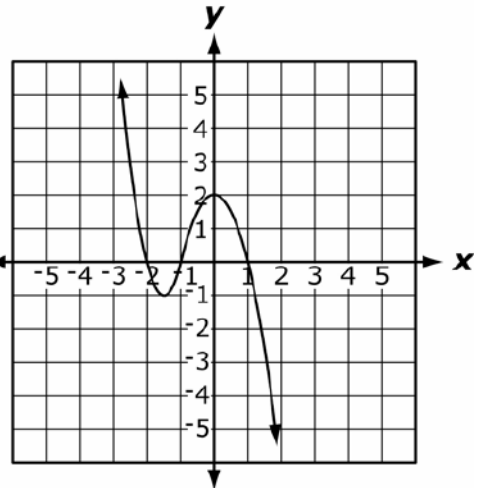
B



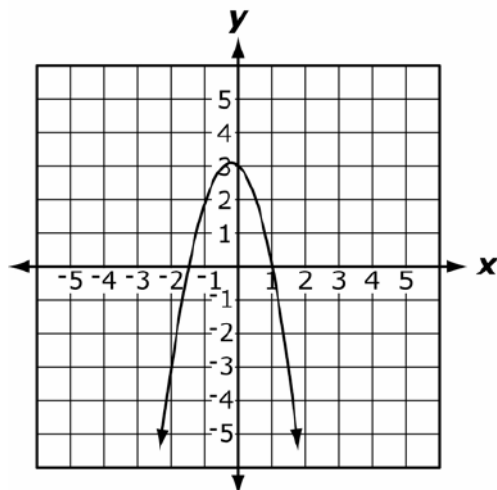
C



D



86. Which function is represented by this graph?



- A $f(x) = 2x^2 - x + 3$
- B $f(x) = -2x^2 - x + 3$
- C $f(x) = 2x^2 - 3x + 1$
- D $f(x) = -2x^2 - 3x + 1$

87. Which statement describes the characteristics of the graph of $f(x) = -5x^4 + 3x^2 + x - 2$?

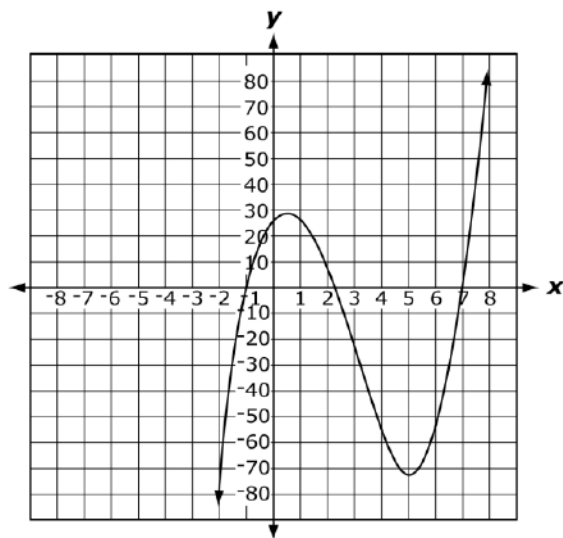
- A The graph primarily increases in the third quadrant and increases in the first quadrant.
- B The graph primarily decreases in the second quadrant and increases in the first quadrant.
- C The graph primarily increases in the third quadrant and decreases in the fourth quadrant.
- D The graph primarily decreases in the second quadrant and decreases in the fourth quadrant.

Objective 2.6c

88. What is the y-intercept of the graph of $y = -4x^2 + 2x - 3$?

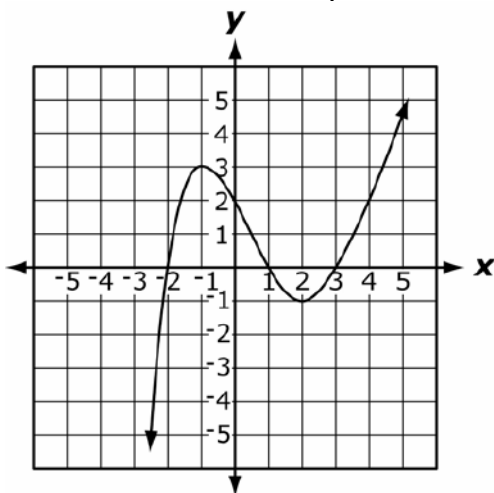
- A -3
- B 3
- C -4
- D 4

89. What are the x- and y-intercepts of this graphed function?



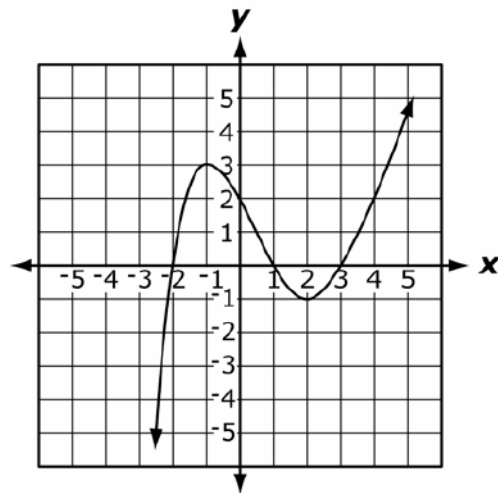
- A x-intercepts: $(-1, 0)$, $(2.3, 0)$, $(7, 0)$; y-intercepts: $(0, 28)$
- B x-intercepts: $(-0, 28)$; y-intercepts: $(-1, 0)$, $(2.3, 0)$, $(7, 0)$
- C x-intercepts: $(1, 0)$, $(-2.3, 0)$, $(-7, 0)$; y-intercepts: $(0, 28)$
- D x-intercepts: $(0, 28)$; y-intercepts: $(1, 0)$, $(-2.3, 0)$, $(-7, 0)$

90. What is the set of x-intercepts of this graphed function?



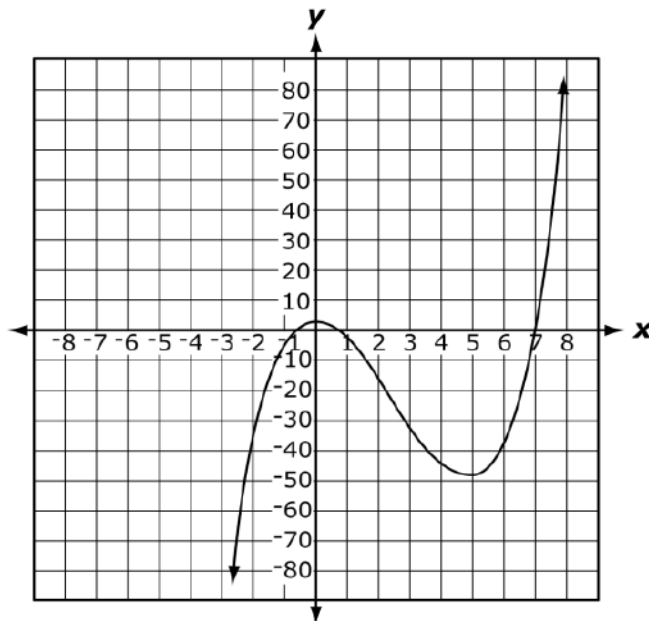
- A $\{2\}$
- B $\{-1, 2\}$
- C $\{-1, 3\}$
- D $\{-2, 1, 3\}$

91. What is the set of approximate y-values of the relative minimum and maximum of this graphed function?



- A $\{2\}$
- B $\{-1, 2\}$
- C $\{-1, 3\}$
- D $\{-2, 1, 3\}$

92. What are the properties of the point $(0, 3)$ in this graphed function?



- A It is a relative minimum and an x-intercept.
- B It is a relative maximum and an x-intercept.
- C It is a relative minimum and an y-intercept.
- D It is a relative maximum and an y-intercept.

Objective 2.6

93. The intensity, L , of light varies inversely with the square of the distance, r , from the source of the light. Given that k is the constant of proportionality, which equation describes this relationship?

A $L = 2kr$
 B $L = \frac{k}{r^2}$
 C $L = k\sqrt{r}$
 D $L = kr^2$

94. A company is selling an item and determines that the profit from selling the item for a price of x dollars is given by the function below.

$$P(x) = \frac{-1}{4}(x-16)^2 + 4$$

Which price will maximize the profit?

- A \$4
 B \$12
 C \$16
 D \$20
95. The path of a kicked soccer ball can be modeled by the function $f(x) = 26 + 2x - x^2$, where x is the horizontal distance (in meters) and $f(x)$ is the height (in meters). If the height is 2 meters, what is the horizontal distance?
- A 4 meters
 B 6 meters
 C 12 meters
 D 24 meters
96. A landscape designer has to construct a rectangular flower bed with a perimeter of 100 feet and the maximum possible area. What is the area of the flower bed?
- A 25 sq. ft
 B 100 sq. ft
 C 625 sq. ft
 D 2,500 sq. ft

Objective 2.7a

97. What is the value of x in this rational equation $\frac{2}{x-1} = \frac{3}{x+1}$?

A 2
 B 3
 C 4
 D 5

98. What is the solution set of this rational equation $\frac{5}{2x-2} - \frac{9}{2x} = \frac{-1}{4}$?

- A {6}
- B {-3}
- C {3, 6}
- D {-3, -6}

99. What is the value of x in this rational equation $2x = \frac{4x+5}{3}$?

- A -4
- B -1
- C $\frac{1}{2}$
- D $\frac{5}{2}$

100. What is the solution set of this rational equation $\frac{-3}{x^2} + \frac{1}{2} = \frac{1}{2x}$?

- A {-3, -2}
- B {-3, 2}
- C {-2, 3}
- D {2, 3}

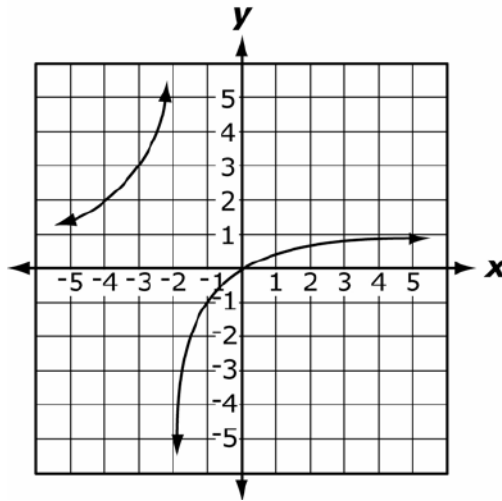
Objective 2.7b

101. What is the vertical asymptote of the graph of $f(x) = \frac{1}{x+4}$?

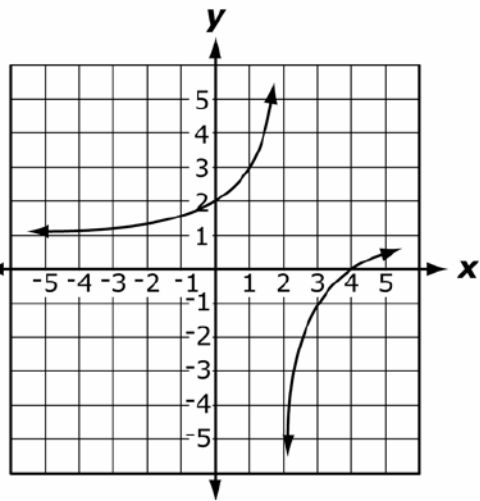
- A $x = -4$
- B $x = -1$
- C $x = 1$
- D $x = 4$

102. What is the graph of the function $f(x) = \frac{x}{x-2}$?

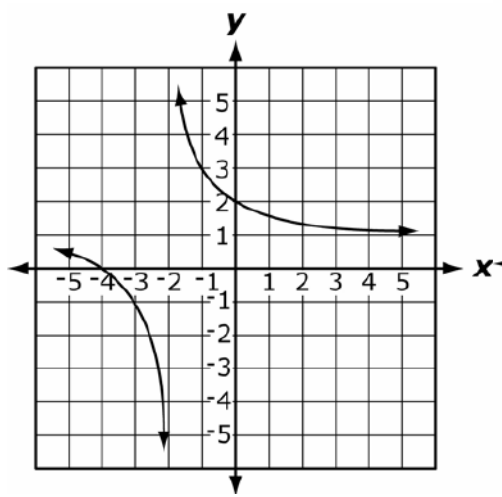
A



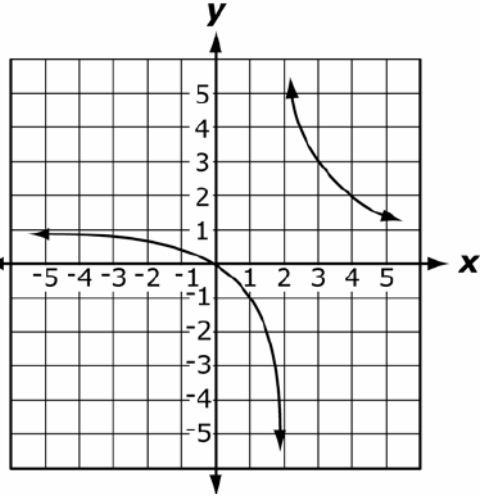
B



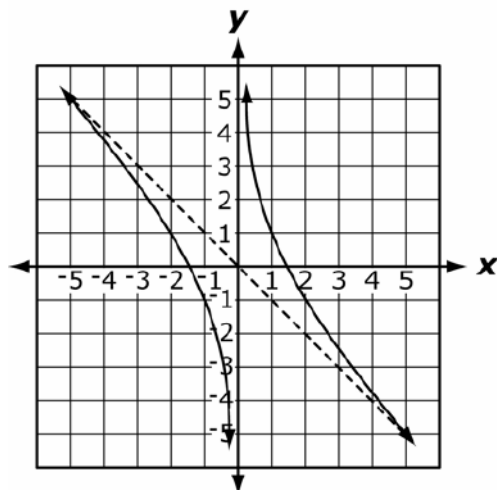
C



D



103. Which function is represented by this graph?

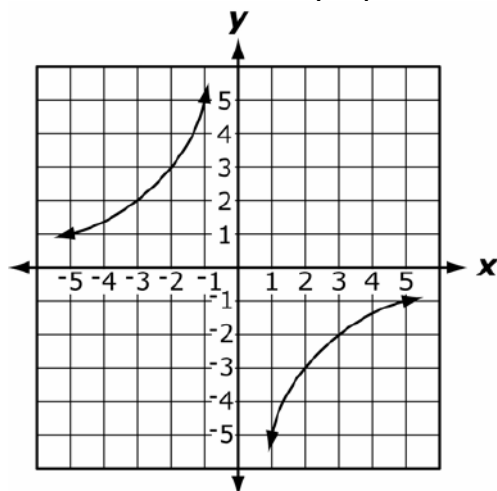


- A $f(x) = \frac{-2 + x^2}{x}$
- B $f(x) = \frac{-2 - x^2}{x}$
- C $f(x) = \frac{2 - x^2}{x}$
- D $f(x) = \frac{2 + x^2}{x}$

104. How many vertical asymptotes does the graph of $y = \frac{x-2}{x^2+4}$ have?

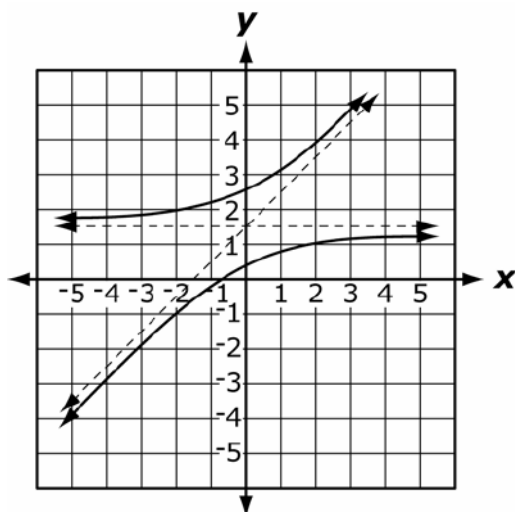
- A 0 vertical asymptotes
- B 1 vertical asymptote
- C 2 vertical asymptotes
- D 4 vertical asymptotes

105. What is the horizontal asymptote of this graph?



- A $x=0$
- B $y=0$
- C $x=2.5$
- D $y=2.5$

106. Which statement correctly describes the asymptotes of the graph of this rational function?



- A The vertical asymptote is $x = \frac{3}{2}$, and there is a negative slant asymptote.
- B The vertical asymptote is $y = \frac{3}{2}$, and there is a negative slant asymptote.
- C The horizontal asymptote is $x = \frac{3}{2}$, and there is a positive slant asymptote.
- D The horizontal asymptote is $y = \frac{3}{2}$, and there is a positive slant asymptote.

107. How many x-intercepts does the graph of $y = \frac{x+1}{x^2-1}$ have?

- A 0
- B 1
- C 2
- D 4

108. What are the vertical and horizontal asymptotes of $f(x) = \frac{x^2-9}{16-x^2}$?

- A $x = \pm 4$, and $y = -1$
- B $y = \pm 4$, and $x = -1$
- C $x = \pm 4$, and $y = 1$
- D $y = \pm 4$, and $x = 1$

Objective 2.7d

109. If the surface area of a closed cylinder is 25 square inches, which equation represents the height of the cylinder in terms of r ?

$$(SA = 2\pi rh + 2\pi r^2)$$

- A $h = \frac{25 + 2\pi r^2}{2\pi r}$
- B $h = \frac{25 - 2\pi r^2}{2\pi r}$
- C $h = 25 + r$
- D $h = 25 - r$

110. A homeowner stocked his pond with fish. The number of fish, F , increases according to the equation, $F = \frac{19(3+2t)}{1+0.05t}$, where t is the time in years. What is the approximate number of fish after 10 years?

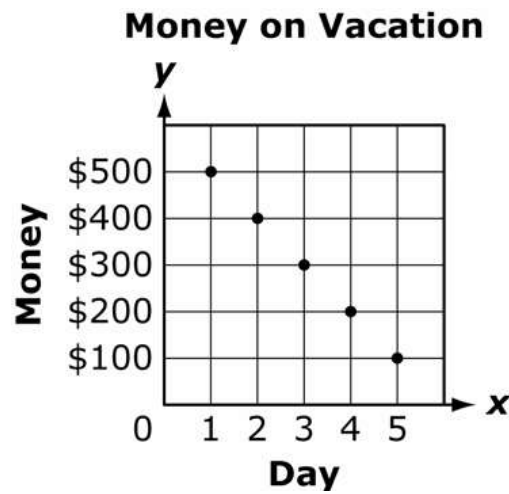
- A 49 fish
- B 69 fish
- C 138 fish
- D 291 fish

111. The cost, C , in thousands of dollars, to remove x percent of the trash left by a tornado is modeled by the equation $C = \frac{450x}{225-x}$. Approximately what percent of trash will be removed if 100 thousand dollars are spent?

- A 41%
- B 50%
- C 59%
- D 64%

Objective 3.1a

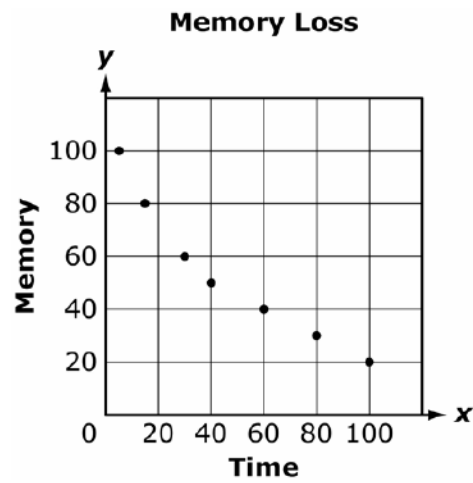
112. Nancy made a scatter plot of how much money she had left at the end of each day of her vacation.



Which table best represents the data in her scatter plot?

- A
- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| Day | 1 | 2 | 3 | 4 | 5 |
| Money | \$100 | \$100 | \$100 | \$100 | \$100 |
- B
- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| Day | 1 | 2 | 3 | 4 | 5 |
| Money | \$100 | \$200 | \$300 | \$400 | \$500 |
- C
- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| Day | 1 | 2 | 3 | 4 | 5 |
| Money | \$500 | \$200 | \$300 | \$400 | \$100 |
- D
- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| Day | 1 | 2 | 3 | 4 | 5 |
| Money | \$500 | \$400 | \$300 | \$200 | \$100 |

113. Which set of data best represents the data on the scatter plot?



A

Time	10	30	60	80	100
Memory	95	60	40	30	20

B

Time	10	30	60	80	100
Memory	20	30	40	60	95

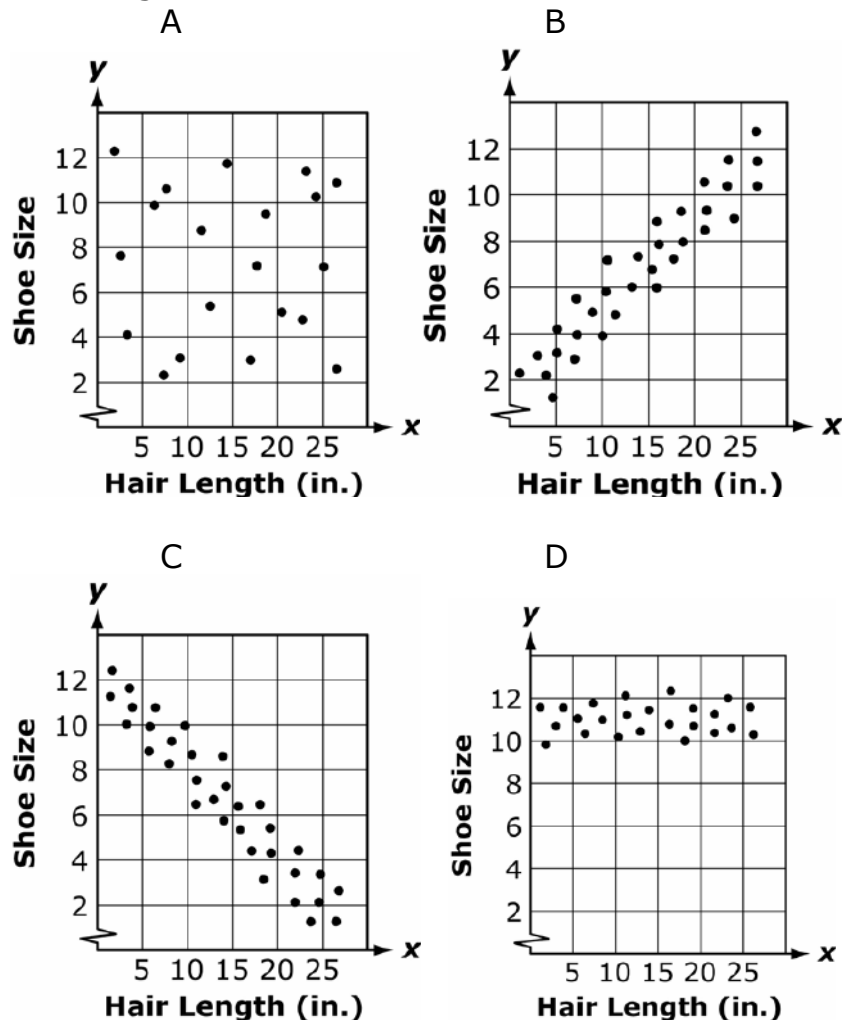
C

Time	10	30	60	80	100
Memory	100	80	60	40	20

D

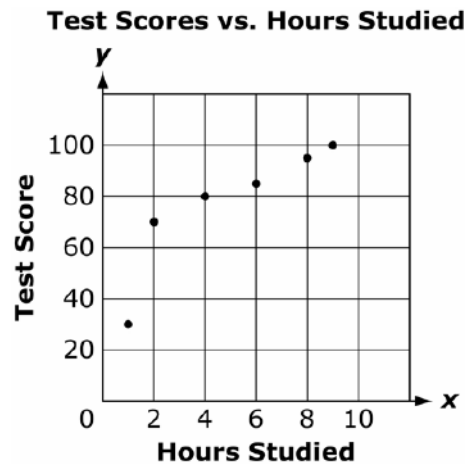
Time	10	30	60	80	100
Memory	85	60	50	40	20

114. Which scatter plot best represents the lack of correlation between shoe size and hair length?



objective 3.1b

115. The test scores and hours studied of 6 students were put into a scatter plot.



If another student studies 2 hours, what is the most likely test score based on this data?

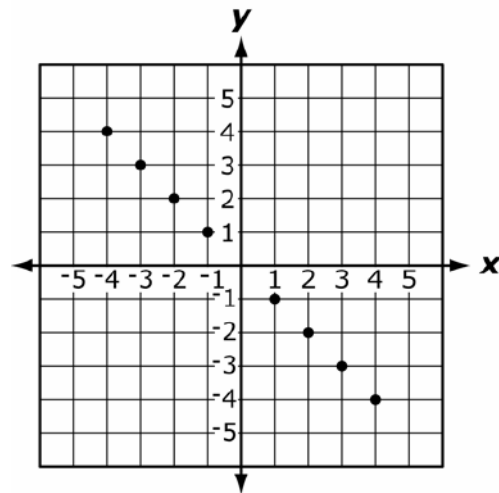
- A 20
- B 60
- C 70
- D 80

116. Which of these observations would be consistent with an exponential model of population growth?

- A The population started out large, decreased in size, then became large again.
- B The population is observed to increase at a faster rate as time passes.
- C The population is observed to increase steadily over time.
- D The population grew very quickly but then declined.

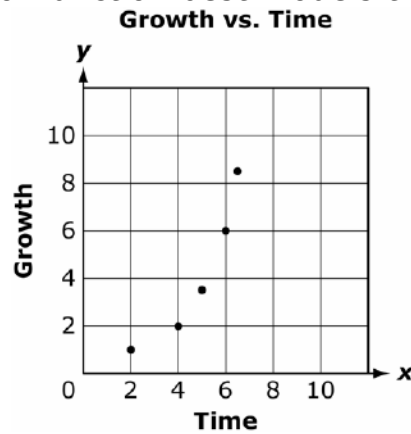
Objective 3.1c

117. Which equation most closely models the data in the scatter plot?



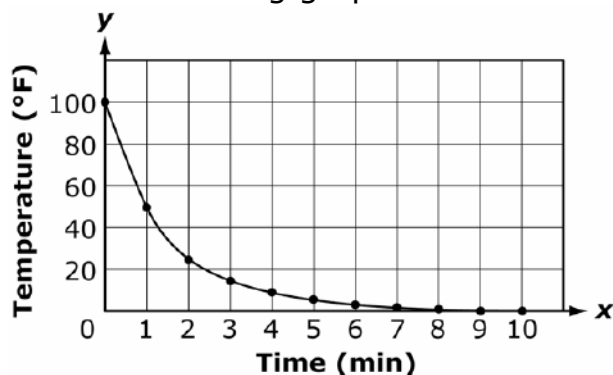
- A $y = x$
- B $y = -x$
- C $y = 2x$
- D $y = -2x$

118. Which type of function best models the data in this scatter plot?



- A exponential
- B logarithmic
- C quadratic
- D linear

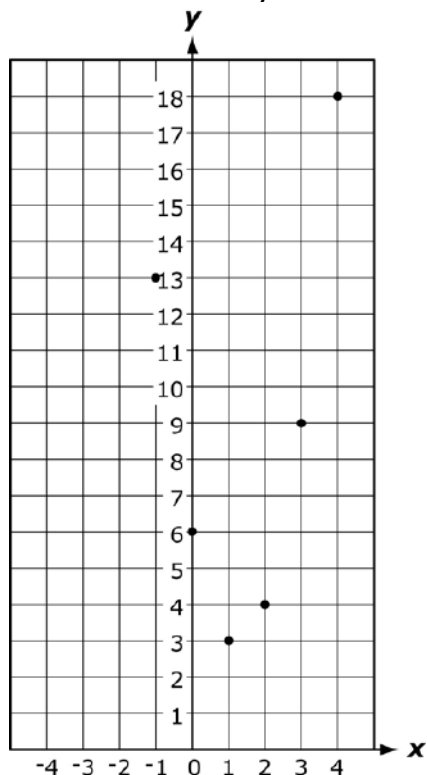
119. Students in a science classroom perform an experiment to find the rate at which a hot liquid cools in a freezer. They plot the temperature over time and obtain the following graph.



Which type of function best models the data in this scatter plot?

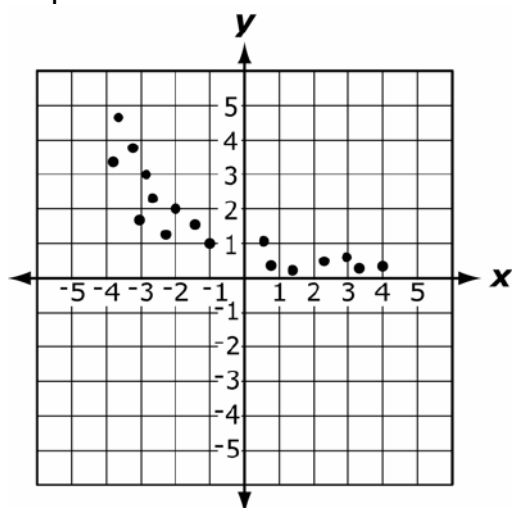
- A exponential
- B logarithmic
- C quadratic
- D linear

120. Which equation most closely models the data in the scatter plot?



- A $y = x^2 - 4x + 6$
- B $y = -x^2 - 2x + 6$
- C $y = -2x^2 - x + 6$
- D $y = 2x^2 - 5x + 6$

121. Which equation best models the data in this scatter plot?



- A $y = 5 \cdot 3^x$
- B $y = 0.5 \cdot 3^x$
- C $y = 5 \cdot 0.5^x$
- D $y = 0.5 \cdot 5^x$

Objective 3.3

122. What is the 12th term in the sequence $\{1, 3, 5, 7, \dots\}$?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A 22
- B 23
- C 24
- D 25

123. What is the sum of the first 6 terms of the series $2+10+50+\dots$?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A 3,906
- B 7,812
- C 15,624
- D 31,248

124. A child puts \$1.00 into a piggy bank. One week later, he puts \$1.25 in the bank. Two weeks later, he puts \$1.50 in the bank, and so on. How much money does he put in the bank on the 25th week?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A \$ 6.25
- B \$7.00
- C \$93.00
- D \$100.00

125. What is the value of x in the geometric sequence $\left\{x, -\frac{1}{2}, \frac{1}{8}, \frac{-1}{32}, \dots\right\}$?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A -4
- B -2
- C 2
- D $\frac{9}{2}$

126. Which formula could be used to find the sum of an arithmetic series if the last term is unknown?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A $s_n = \frac{n}{2}(2a_1 + (n-1)d)$
- B $s_n = \frac{n}{2}(2a_1 + (n+1)d)$
- C $s_n = n(2a_1 + (n-1)d)$
- D $s_n = n(2a_1 + (n+1)d)$

127. In an arithmetic sequence beginning with 36 and ending with 405, how many integers are divisible by 9?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A 41 integers
- B 42 integers
- C 44 integers
- D 45 integers

128. How many terms are there in a geometric series if the first term is 3, the common ratio is 4, and the sum of the series is 1,023?

Arithmetic Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 + (n-1)d$$

$$\text{Sum: } s_n = \frac{n}{2}(a_1 + a_n)$$

Geometric Sequences & Series

$$n^{\text{th}} \text{ term: } a_n = a_1 r^{(n-1)}$$

$$\text{Sum: } s_n = \frac{a_1(1-r^n)}{(1-r)}$$

- A 4 terms
- B 5 terms
- C 6 terms
- D 23 terms

Answers

1. D
2. B
3. A
4. A
5. B
6. B
7. B
8. A
9. C
10. A
11. A
12. B
13. D
14. C
15. C
16. D
17. D
18. D
19. C
20. D
21. A
22. B
23. B
24. D
25. A
26. D
27. D
28. D
29. C
30. B
31. D
32. B
33. C
34. B
35. A
36. A
37. C
38. A
39. A
40. B
41. B
42. C
43. B
44. D

45. A
46. B
47. B
48. B
49. C
50. A
51. D
52. B
53. B
54. A
55. C
56. B
57. D
58. A
59. B
60. D
61. B
62. A
63. B
64. C
65. A
66. A
67. A
68. C
69. A
70. B
71. B
72. B
73. D
74. B
75. B
76. A
77. C
78. A
79. B
80. B
81. A
82. A
83. A
84. B
85. C
86. B
87. C
88. A

89. A
90. D
91. C
92. D
93. B
94. C
95. B
96. C
97. D
98. C
99. D
100. C
101. A
102. D
103. C
104. A
105. B
106. D
107. A
108. A
109. B
110. D
111. A
112. D
113. A
114. A
115. C
116. B
117. B
118. A
119. A
120. D
121. C
122. B
123. B
124. B
125. C
126. A
127. A
128. B

Student Name _____

P



Algebra II
Test Booklet

Practice Test
Large Print

TEST BOOKLET SECURITY BARCODE

Section 1

(Non-Calculator)

Directions:

Today, you will take Section 1 of the Algebra II Practice Test. You will not be able to use a calculator.

Read each question. Then, follow the directions to answer each question. Circle the answer or answers you have chosen in your test booklet. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses entered within the space provided will be scored.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this section ONLY. Do not go past the stop sign.

Directions for Completing the Answer Grids

1. Work the problem and find an answer.
2. Write your answer in the boxes at the top of the grid.
3. Print only one number or symbol in each box. Do not leave a blank box in the middle of an answer.
4. Fractions cannot be entered into an answer grid and will not be scored. Enter fractions as decimals.
5. See below for examples on how to correctly complete an answer grid.

EXAMPLES

To answer -3 in a question, fill in the answer grid as shown below.

-	3					
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

To answer $.75$ in a question, fill in the answer grid as shown below.

.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

- 1 Which number is equivalent to $(5 + i)^2$, where i is the imaginary unit?

Select one answer.

- A 24
- B $24 + 10i$
- C 26
- D $26 + 10i$

- 2 What are the solutions of the equation $\frac{x}{x+6} = \frac{2x-3}{x+12}$?

Select **all** that apply.

- A $x = -12$
- B $x = -6$
- C $x = -3$
- D $x = 0$
- E $x = \frac{3}{2}$
- F $x = 3$
- G $x = 6$

- 3 The recursive rule for a sequence is shown, where n represents a positive integer.

$$\begin{aligned} f(1) &= 5 \\ f(n) &= 3 \cdot f(n - 1), \text{ for } n > 1 \end{aligned}$$

Which is an explicit rule for the sequence?

- A** $f(n) = 3(5)^{n-1}$, for $n \geq 1$
- B** $f(n) = 5(3)^{n-1}$, for $n \geq 1$
- C** $f(n) = 3 + 5(n - 1)$, for $n \geq 1$
- D** $f(n) = 5 + 3(n - 1)$, for $n \geq 1$
- 4 The binomial $(x - 2)$ is a factor of the polynomial function $P(x) = x^3 - x^2 + ax - 6$, where a is a constant.

What is the value of a ?

Enter your answer in the space provided.

⊖							
	⊙	⊙	⊙	⊙	⊙	⊙	⊙

- 5 Function f is defined as $f(x) = (0.5)^{2x}$.

Which statement correctly describes how the value of $f(x)$ is changing for every increase of 1 in the value of x ?

Select one answer.

- A** The value of $f(x)$ increases by 25%.
- B** The value of $f(x)$ decreases by 25%.
- C** The value of $f(x)$ increases by 75%.
- D** The value of $f(x)$ decreases by 75%.

- 6 An expression is shown.

$$\left(\frac{64}{27}\right)^{-\frac{1}{3}}$$

Using the properties of exponents, the expression can be rewritten as which of the fractions shown?

A $-\frac{4}{3}$

B $-\frac{3}{4}$

C $\frac{3}{4}$

D $\frac{4}{3}$

- 7 The function f is defined by $f(x) = 2x + 5$.

If $f(a) = c$, which of the following equations gives the value of a in terms of c ?

Select one answer.

A $a = 2c + 5$

B $a = 5c + 2$

C $a = \frac{c - 5}{2}$

D $a = \frac{c}{2} - 5$

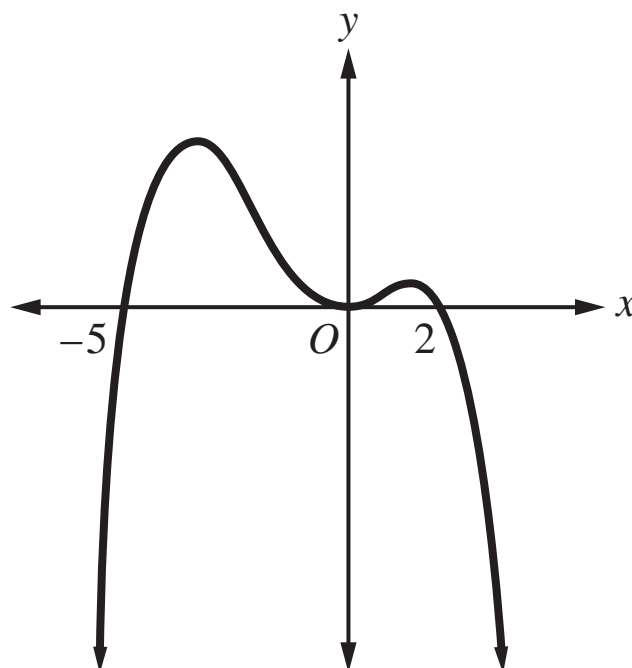
- 8 The expression $x^3 - 3x^2 - 4x + 12$ can be written as $(x^2 - a)(x - b)$, where a and b are constants.

What are the values of a and b ?

- A** $a = 1$ and $b = 12$
- B** $a = 2$ and $b = 6$
- C** $a = 3$ and $b = 4$
- D** $a = 4$ and $b = 3$
- 9 Which value of x makes the equation $5(e)^{4x} = 25$ true?

- A** $x = \ln \frac{5}{4}$
- B** $x = \frac{1}{4} \ln 2$
- C** $x = \frac{1}{4} \ln 5$
- D** $x = \frac{1}{20} \ln 25$

- 10** The graph of a polynomial function is shown in the following xy -plane.

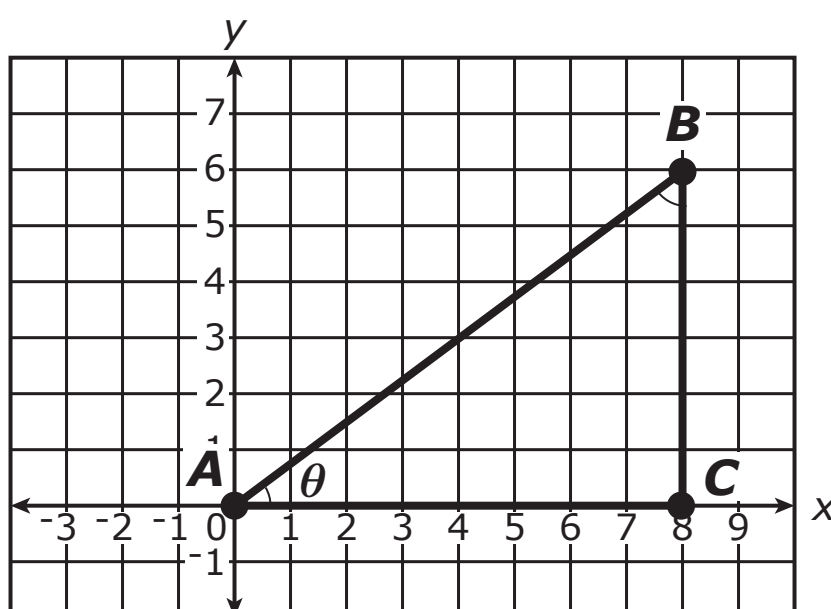


Which equation could represent the graph of the polynomial function?

Select one answer.

- A** $y = x^2(x + 5)(x - 2)$
- B** $y = -x^2(x + 5)(x - 2)$
- C** $y = x^2(x - 5)(x + 2)$
- D** $y = -x^2(x - 5)(x + 2)$

- 11** Right triangle ABC is shown in the xy -coordinate plane. Point B has coordinates $(8, 6)$.



Which statements are true?

Select **all** that apply.

A $\sin \theta = -\frac{4}{5}$

B $\sin \theta = \frac{3}{5}$

C $\sin \theta = \frac{4}{5}$

D $\sin(\pi + \theta) = -\frac{4}{5}$

E $\sin(\pi + \theta) = -\frac{3}{5}$

F $\sin(\pi + \theta) = \frac{3}{5}$

12 Which table shows the three quantities 4^7 , $(4^6)(4^2)$, and $9^{3.5}$ in order from **least** to **greatest**?

A

Least	$9^{3.5}$
	$(4^6)(4^2)$
Greatest	4^7

B

Least	$9^{3.5}$
	4^7
Greatest	$(4^6)(4^2)$

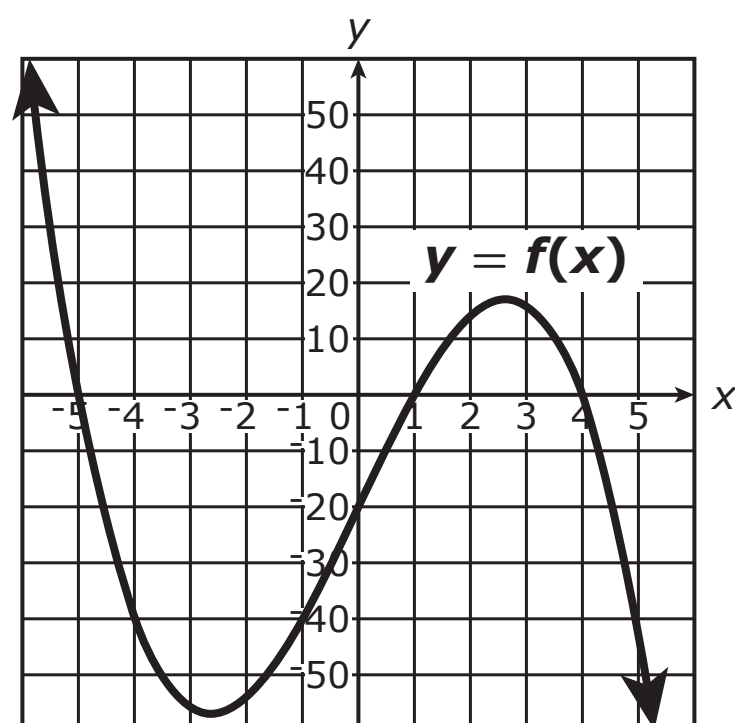
C

Least	4^7
	$9^{3.5}$
Greatest	$(4^6)(4^2)$

D

Least	4^7
	$(4^6)(4^2)$
Greatest	$9^{3.5}$

- 13** The graph of the polynomial function f is shown in the xy -coordinate plane.



Which of the equations shown could define f ?

- A** $f(x) = -(x + 5)(x - 1)(x - 4)$
- B** $f(x) = (x + 5)(x - 1)(x - 4)$
- C** $f(x) = -(x - 5)(x + 1)(x + 4)$
- D** $f(x) = (x - 5)(x + 1)(x + 4)$



You have come to the end of Section 1 of the test. Review your answers from Section 1 only.





GO ON TO NEXT PAGE



Section 2

(Calculator)

Directions:

Today, you will take Section 2 of the Algebra II Practice Test. You will be able to use a calculator.

Read each question. Then, follow the directions to answer each question. Circle the answer or answers you have chosen in your test booklet. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses entered within the space provided will be scored.

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4. Fractions cannot be entered into an answer grid and will not be scored. Enter fractions as decimals.
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EXAMPLES

To answer -3 in a question, fill in the answer grid as shown below.

-	3					
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

To answer $.75$ in a question, fill in the answer grid as shown below.

.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



Mathematics

Section 2

- 1 The table shows the percent of US households with computers for selected years between 1984 and 2003.

Percent of Households with Computers

Year	Percent
1984	8.2
1989	15.0
1993	22.9
1997	36.6
2000	51.0
2001	56.3
2003	61.8

Which of the equations shown **best** models the data in the table, where y is the percent of households with computers and x is the number of years after 1980?

A $y = 5.6(1.08)^x$

B $y = 5.6(1.09)^x$

C $y = 5.6(1.11)^x$

D $y = 5.6(1.14)^x$



2 The functions f and g are defined as shown.

$$f(x) = x^3 + x^2 - 2x$$

$$g(x) = 0.5x^2 + 1$$

The values of x for which $f(x) = g(x)$ lie in which of the intervals shown?

Select **all** that apply.

A $-3 < x < -2$

B $-2 < x \leq -1$

C $-1 < x \leq 0$

D $0 < x \leq 1$

E $1 < x \leq 2$

F $2 < x < 3$



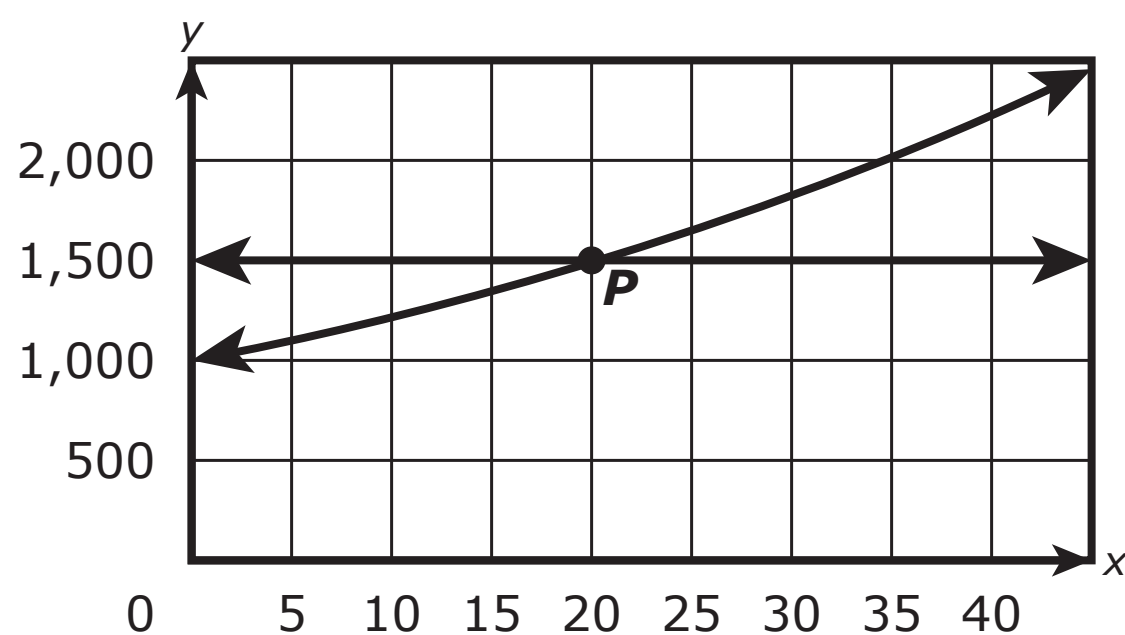
Mathematics

Section 2

- 3** An account that pays 0.5% interest was opened with an initial deposit, and no additional deposits to or withdrawals from the account were made. The equation $y = 1000(1.005)^{4x}$ models the amount, in dollars, in the account x years after it was opened.

The graph of the model and the graph of the equation $y = 1500$ are shown in the xy -coordinate plane.

The intersection of the two graphs is marked with the point P .



What is revealed by the coordinates of the point P ?

- A** the yearly interest rate for the account
- B** the amount of the initial deposit in the account
- C** the number of years it takes for the account value to reach \$1500
- D** the number of years it takes for the initial amount in the account to double



GO ON TO NEXT PAGE



Mathematics

- 4 Claim: There is no quadratic equation with real coefficients for which $x = -5i$ is a solution.

Part A

Show that the claim is not correct by providing an example of a quadratic equation for which $x = -5i$ is a solution. Include any other solutions to your equation in your answer.

Enter your answer and your explanation in the space provided.

Section 2

4 Part A

**Part B**

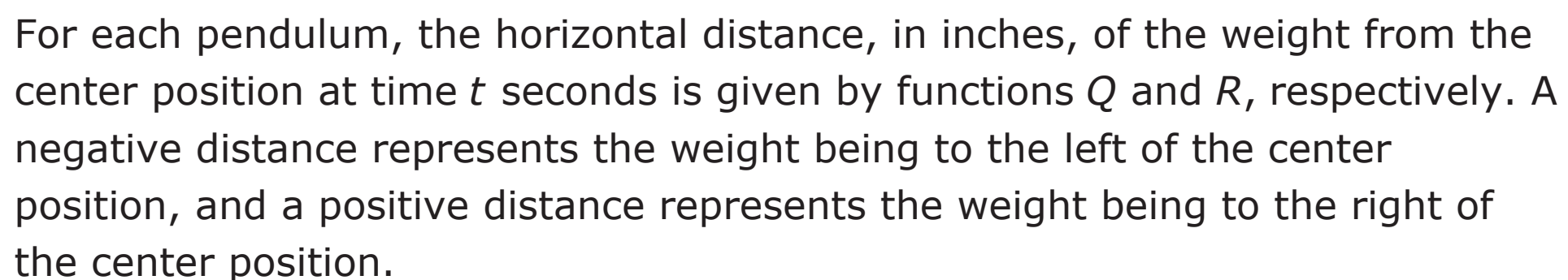
Is it possible for a quadratic equation with real coefficients to have $x = -5i$ as its **only** solution? Justify your answer.

Enter your answer and your justification in the space provided.

4 Part B**Section 2**



5 For the two pendulums shown, a weight swings back and forth from left to right. Both pendulums start at the left position at time $t = 0$ seconds.



$$\begin{aligned} Q(t) &= -6\cos(\pi t) \\ R(t) &= -4\cos\left(\frac{4}{3}\pi t\right) \end{aligned}$$

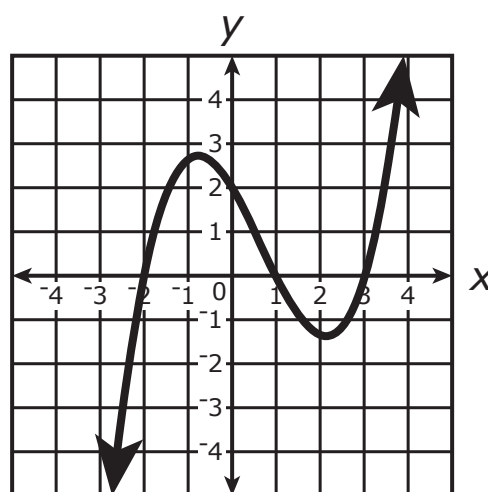
The maximum horizontal distance of pendulum Q to the right of its starting position is how much greater, in inches, than the maximum horizontal distance of pendulum R to the right of its starting position?

Enter your answer in the space provided.

⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



- 6 Consider the polynomial function P , defined by $P(x) = x^3 + cx^2 + x + 2$, where c is a constant, and consider the graph shown in the xy -coordinate plane.



If $P(-2) = 0$, can the graph shown be the graph of $y = P(x)$?

- A** Yes, because c would equal -2 , and the points $(1, 0)$ and $(3, 0)$ would be on the graph of $y = P(x)$.
- B** Yes, because c would equal 2 , and the points $(1, 0)$ and $(3, 0)$ would be on the graph of $y = P(x)$.
- C** No, because c would equal -2 , and the points $(-2, 0)$ and $(3, 0)$ would not be on the graph of $y = P(x)$.
- D** No, because c would equal 2 , and the points $(1, 0)$ and $(3, 0)$ would not be on the graph of $y = P(x)$.



Mathematics

Section 2

- 7** When a customer pays with a debit card at a clothing store, the owner pays a processing fee. The function P represents the processing fee, in dollars, for a debit card purchase of x dollars.

$$P(x) = 0.035x + 0.15$$

How is the processing fee calculated based on the purchase amount?

- A** The processing fee consists of a fixed amount of \$0.15 plus 0.35% of the purchase amount.
- B** The processing fee consists of a fixed amount of \$0.15 plus 3.5% of the purchase amount.
- C** The processing fee consists of a fixed amount of \$0.35 plus 1.5% of the purchase amount.
- D** The processing fee consists of a fixed amount of \$0.35 plus 15% of the purchase amount.

- 8** An equation is shown.

$$\sqrt{x^2 + 19} - 100 = 0$$

What are the solutions of the equation?

- A** $x = 9$ only
- B** $x = 3\sqrt{1109}$ only
- C** $x = -9$ and $x = 9$
- D** $x = -3\sqrt{1109}$ and $x = 3\sqrt{1109}$





You have come to the end of Section 2 of the test. Review your answers from Section 2 only.





Section 3

(Calculator)

Directions:

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EXAMPLES

To answer -3 in a question, fill in the answer grid as shown below.

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⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

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.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



Mathematics

- 1 The function C approximates the number of calories recommended per day for a dog, based on the weight, in pounds, of the dog.

The table gives values of C for selected values of p , where $C(p)$ represents the recommended number of calories per day for a dog that weighs p pounds.

Recommended Number of Calories for Dogs

p (pounds)	$C(p)$
11	250
22	450
33	750
44	1000
55	1250
66	1500
77	1700
88	1880

Which of the equations shown **most** closely approximates C ?

- A** $C(x) = 18p + 22$
- B** $C(x) = 22p + 18$
- C** $C(x) = 40(1.06)^p$
- D** $C(x) = 48(1.07)^p$



2 Claim: $\sqrt[3]{x} \leq x$, where x is a real number

For which intervals is the claim true for all values of x in the interval?

Select **all** that apply.

A $x > 1$

B $0 < x < 1$

C $-1 < x < 0$

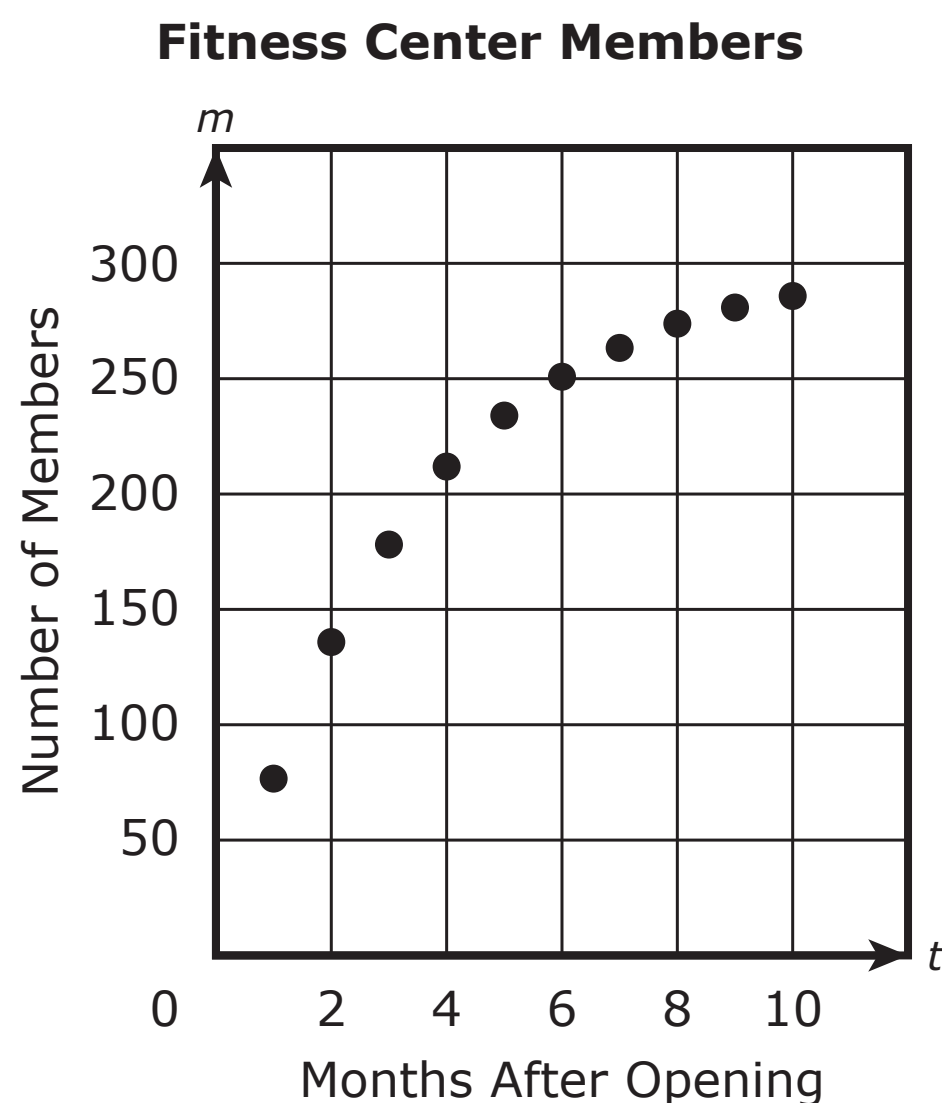
D $-2 < x < -1$

E $x < -2$



Mathematics

- 3 The number of members, m , that a new fitness center expects to have t months after the opening of the center is shown in the scatter plot.



The fitness center constructed the model $m(t) = 300 - 280(0.76)^t$ to project the number of members that can be accommodated t months after opening.

- What is the average rate of change, in members per month, in the number of members the fitness center expects to have from the 4th to the 8th month after opening? Show your work.
- What is the significance of the number 300 in the model? Explain your answer.

Enter your answers and your work and explanation in the space provided.



3

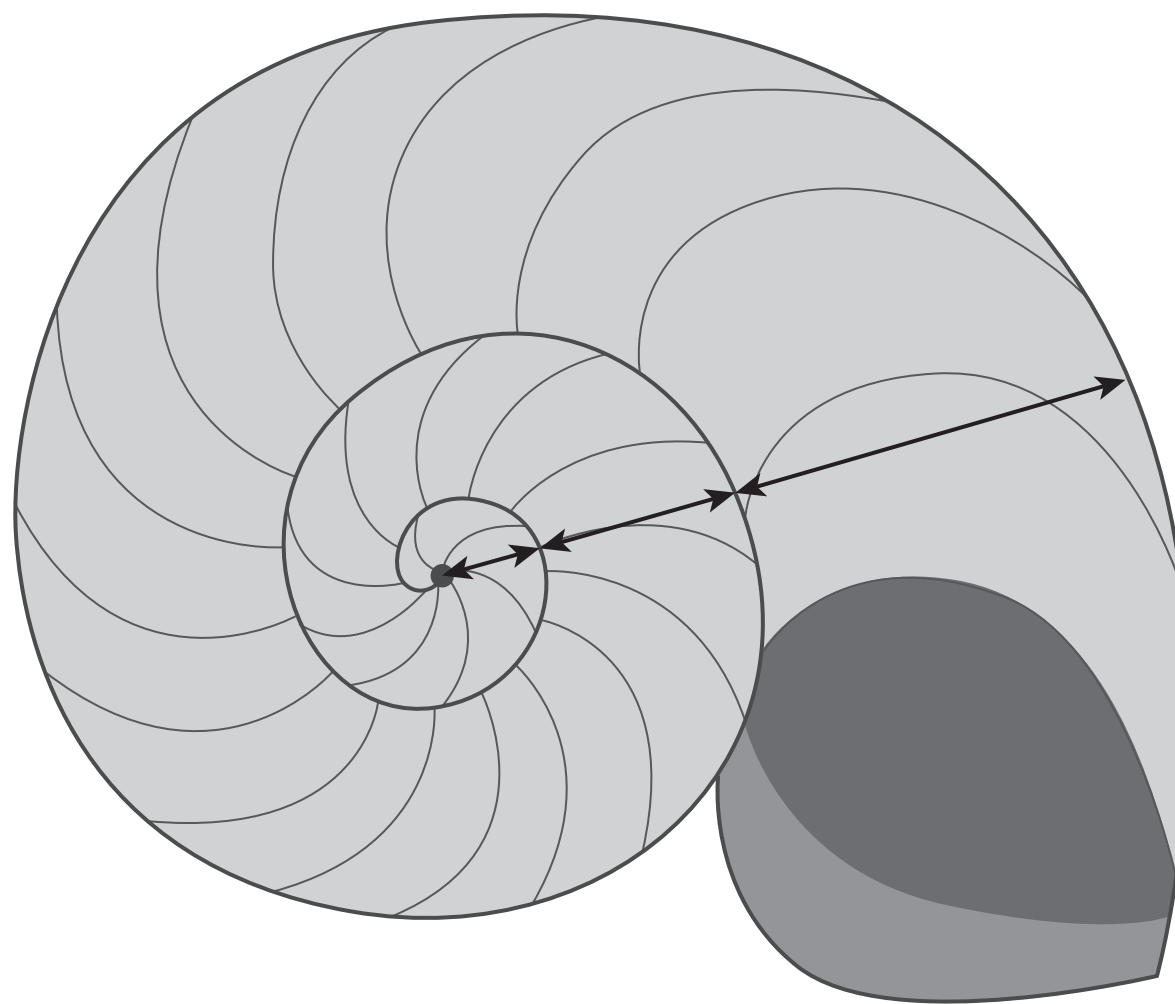
Section 3



Mathematics

- 4 The students in a class are analyzing several examples of spirals in nature.

The students measured the distances shown in the picture for different shells.



Section 3

Each student measured the three distances shown in the picture by starting from the outside of the shell and moving toward the center.

One student's measurements were 8.1 centimeters, 4.2 centimeters, and 2.1 centimeters. A second student's measurements of a different shell were 7.6 centimeters, 3.7 centimeters, and 1.8 centimeters.

Which equation could be used to model the students' data, where a represents the first measurement, in centimeters, and d represents the n th measurement, in centimeters, where $n = 1, 2$, or 3 ?

A $d = a\left(\frac{1}{2}\right)^{n-1}$

B $d = a(2^{n-1})$

C $d = a\left(\frac{1}{2}\right)^n$

D $d = a(2^n)$



GO ON TO NEXT PAGE

Section 3



Mathematics

- 5 Consider the functions P and Q , defined as shown.

$$P(x) = x^2 + 7x - 14$$

$$Q(x) = -3x + 10$$

In the xy -coordinate plane, what are the coordinates of the points at which the graphs of the equations $y = P(x)$ and $y = Q(x)$ intersect?

Explain how you determined your answer.

Enter your answer and your explanation in the space provided.



5



Mathematics

- 6** The approximate number of cars parked in a garage x hours after 6:00 a.m. can be modeled by the function $y = 180 + 130\sin(\frac{x}{12}\pi)$, where $0 \leq x \leq 16$.

Based on the model, for which time interval did the number of cars parked in the garage increase?

- A** 7:00 a.m. to 9:00 a.m.
- B** 1:00 p.m. to 4:00 p.m.
- C** 5:00 p.m. to 7:00 p.m.
- D** 8:00 p.m. to 10:00 p.m.





Section 3

You have come to the end of Section 3 of the test. Review your answers from Section 3 only.





Section 4

(Calculator)

Directions:

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To answer $.75$ in a question, fill in the answer grid as shown below.

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⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

**Mathematics**

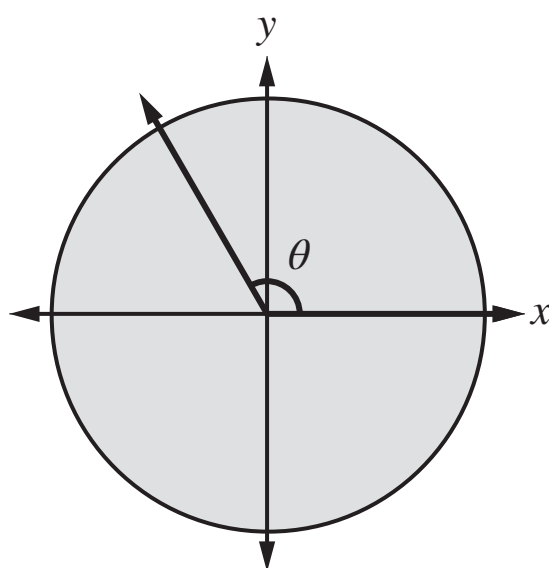
- 1** The half-life of the isotope Francium-223 is 22 minutes. For a sample that starts with F grams of Francium-223, the expression $F(0.5)^{\frac{m}{22}}$ represents the number of grams remaining after m minutes.

Which expression represents the number of grams of Francium-223 remaining after h hours?

- A** $F(0.151)^h$
- B** $F(0.5)^{\frac{h}{60}}$
- C** $F(0.5)^{60h}$
- D** $F(0.969)^h$



- 2 Angle θ is positioned in a circle with its initial ray on the positive x -axis and its terminal ray $\frac{1}{3}$ of one complete revolution about the origin, as shown in the following diagram.



Which of the following statements correctly expresses the measure of θ in both degrees and radians?

Select one answer.

- A** The measure of angle θ is 120 degrees or $\frac{\pi}{3}$ radians.
- B** The measure of angle θ is 120 degrees or $\frac{2\pi}{3}$ radians.
- C** The measure of angle θ is 150 degrees or $\frac{5\pi}{12}$ radians.
- D** The measure of angle θ is 150 degrees or $\frac{7\pi}{12}$ radians.
- 3 Claim: For every value of the constant c , the equation $x^4 + (1 - c^2)x^2 - c^2 = 0$ has at least two distinct real solutions.

What value of c refutes the claim?

Enter your answer in the space provided.

⊖							
⊙	⊙	⊙	⊙	⊙	⊙	⊙	⊙



- 4 The table shows the box office revenue, in millions of dollars, for the first 10 weeks after a movie opened in theaters.

Box Office Revenue for a Movie

Number of Weeks after Opening	Box Office Revenue (in millions)
0	\$85.63
1	\$51.97
2	\$37.21
3	\$25.20
4	\$15.22
5	\$11.17
6	\$7.02
7	\$4.03
8	\$4.42
9	\$1.95

**Part A**

Create a function to model the data. Explain why you chose the type of function and explain the meaning of the parameters of the function.

Enter your answer and your explanation in the space provided.

4 Part A



Mathematics

Part B

The theaters will stop showing the movie when the weekly box office revenue is below \$100,000. In what week does your model predict that the theaters will stop showing the movie?

Show your work or explain how you found your answer.

Enter your answer and your work or explanation in the space provided.

4 Part B



- 5 Consider the equation $a = \sqrt{x + b}$, where a and b represent real numbers.

Which statement is true about the number of real solutions to the equation?

- A The number of real solutions depends only on the value of a .
- B The number of real solutions depends only on the value of b .
- C The number of real solutions depends on the relationship between a and b .
- D The values of a and b do not affect the number of real solutions.

- 6 The world population in 1997 was 5.88 billion.

The world population in 2017 was 7.53 billion.

Assume that the ratio between the population in two consecutive years was constant between 1997 and 2017.

Which equation can be used to find r , the rate of growth per year of the world population?

- A $5.88 = 7.53r^{10}$
- B $5.88 = 7.53r^{20}$
- C $7.53 = 5.88r^{10}$
- D $7.53 = 5.88r^{20}$



Mathematics

7 Consider the system of equations shown.

$$y = (x - 3)^2$$
$$y = x - 1$$

On the coordinate plane, what are the coordinates of the points that represent solutions (x, y) of the system?

A $(-5, -6)$ and $(-2, -3)$

B $(-5, -6)$ and $(2, 1)$

C $(5, 4)$ and $(-2, -3)$

D $(5, 4)$ and $(2, 1)$

8 A kitchen supply store sells a set of 4 food storage containers.

- From largest to smallest, the volumes of the containers are v_1, v_2, v_3 , and v_4 .
- v_n is 20% less than v_{n-1} , where n is 2, 3, or 4.
- $v_n = kv_{n-1}$, where n is 2, 3, or 4.

What is the value of k ?

A $\frac{3}{4}$

B $\frac{4}{5}$

C $\frac{5}{4}$

D $\frac{4}{3}$





You have come to the end of Section 4 of the test. Review your answers from Section 4 only.

Section 4



P

ALG II